

Rookery Bay National Estuarine Research Reserve

SEACAR Habitat Analyses

Last compiled on 10 June, 2026

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Funding & Acknowledgements

The data used in this analysis is from the Export Standardized Tables in the SEACAR Data Discovery Interface (DDI). Documents and information available through the SEACAR DDI are owned by the data provider(s) and users are expected to provide appropriate credit following accepted citation formats. Users are encouraged to access data to maximize utilization of gained knowledge, reducing redundant research and facilitating partnerships and scientific innovation.

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Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR_QAQCFlagCode* and *SEACAR_QAQC_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Dissolved Oxygen	mg/L	-0.000001	50
Dissolved Oxygen Saturation	%	-0.000001	500
Fluorescent Dissolved Organic Matter	QSE	-	-
Salinity	ppt	-0.000001	70
Specific Conductivity	mS/cm	-0.000001	200
Turbidity	NTU	-0.000001	4000
Water Temperature	Degrees C	-5.000000	45
pH	None	2.000000	14

Table 2: Discrete Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Ammonia, Un-ionized (NH3)	mg/L	-	-
Ammonium (NH4)	mg/L	-	-

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Corrected for Pheophytin	ug/L	-	-
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Colored Dissolved Organic Matter	PCU	-	-
Dissolved Oxygen	mg/L	-0.000001	25
Dissolved Oxygen Saturation	%	-0.000001	310
Fluorescent Dissolved Organic Matter	QSE	-	-
Light Extinction Coefficient	m ⁻¹	-	-
NO2+3, Filtered	mg/L	-	-
Nitrate (NO3)	mg/L	-	-
Nitrite (NO2)	mg/L	-	-
Nitrogen, inorganic	mg/L	-	-
Nitrogen, organic	mg/L	-	-
Phosphate, Filtered (PO4)	mg/L	-	-
Salinity	ppt	-0.000001	70
Secchi Depth	m	0.000001	50
Specific Conductivity	mS/cm	0.005000	100
Total Ammonia (N)	mg/L	-	-
Total Kjeldahl Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Phosphorus	mg/L	-	-
Total Suspended Solids	mg/L	-	-
Turbidity	NTU	-	-
Water Temperature	Degrees C	3.000000	40
pH	None	2.000000	13

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

<i>SEACAR QAQC Description</i>	<i>Include</i>	<i>SEACAR QAQCFlagCode</i>
Exceeds maximum threshold	0	2Q
Below minimum threshold	0	4Q
Within threshold tolerance	1	6Q
No defined thresholds for this parameter	1	7Q

Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>MDL</i>	<i>Description</i>
STORET-WIN	H	0	0	Value based on field kit determination; results may not be accurate
STORET-WIN	J	0	0	Estimated value
STORET-WIN	V	0	0	Analyte was detected at or above method detection limit
STORET-WIN	Y	0	0	Lab analysis from an improperly preserved sample; data may be inaccurate

Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476** - *Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network* and data flagged with Value Qualifier **H** are included for this program only.

H - Value based on field kit determination; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

I - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

Q - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

S - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

U - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>Description</i>
SWMP	-1	1	Optional parameter not collected
SWMP	-2	0	Missing data
SWMP	-3	0	Data rejected due to QA/QC
SWMP	-4	0	Outside low sensor range
SWMP	-5	0	Outside high sensor range
SWMP	0	1	Passed initial QA/QC checks
SWMP	1	0	Suspect data
SWMP	2	1	Reserved for future use
SWMP	3	1	Calculated data: non-vented depth/level sensor correction for changes in barometric pressure
SWMP	4	1	Historical: Pre-auto QA/QC
SWMP	5	1	Corrected data

Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as [El Niño/La Niña](#), large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as "insufficient data to conduct analysis". Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as "Significant Trend" (when $p < 0.05$), or "Non-significant Trend" (when $p \geq 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-May-18.txt*
- *Combined_WQ_WC_NUT_pH-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-May-18.txt*

Chlorophyll a, Corrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

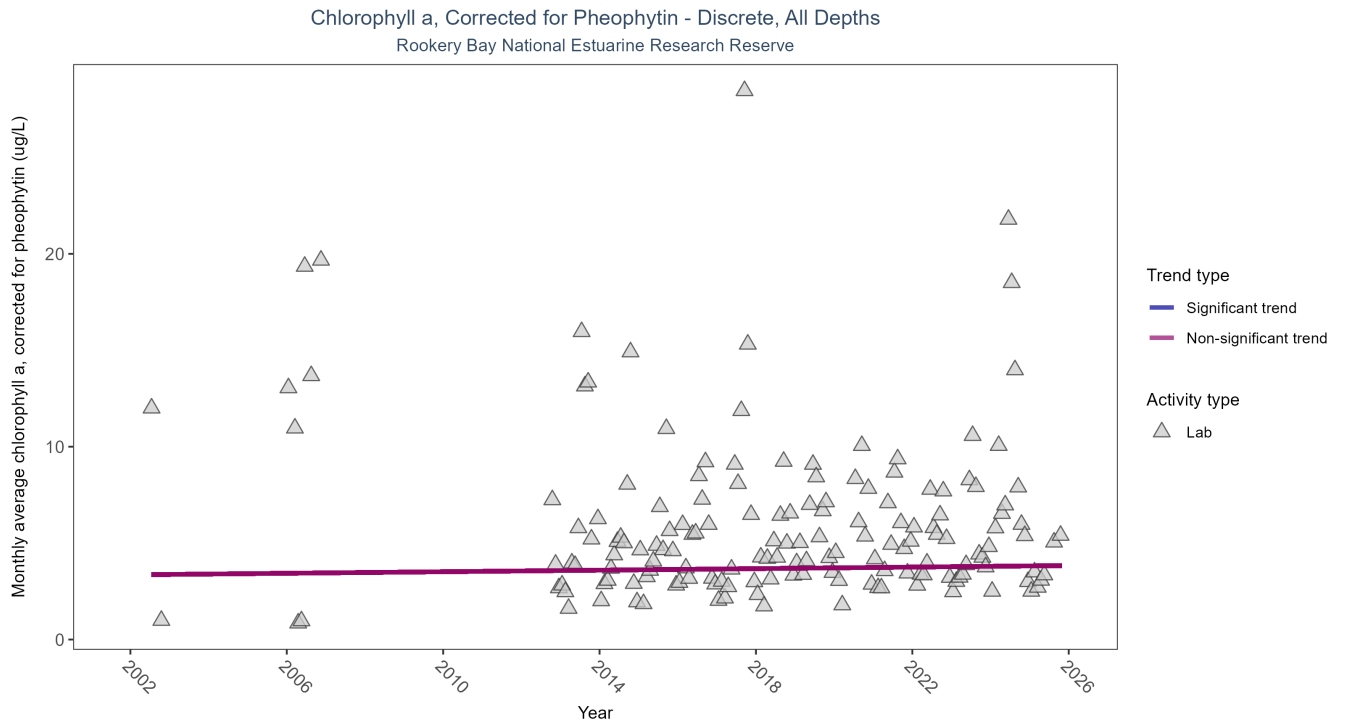


Figure 1: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	2391	16	2002 - 2025	4.1	0.046	3.3559	0.02	0.4535

Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2002 and 2025.

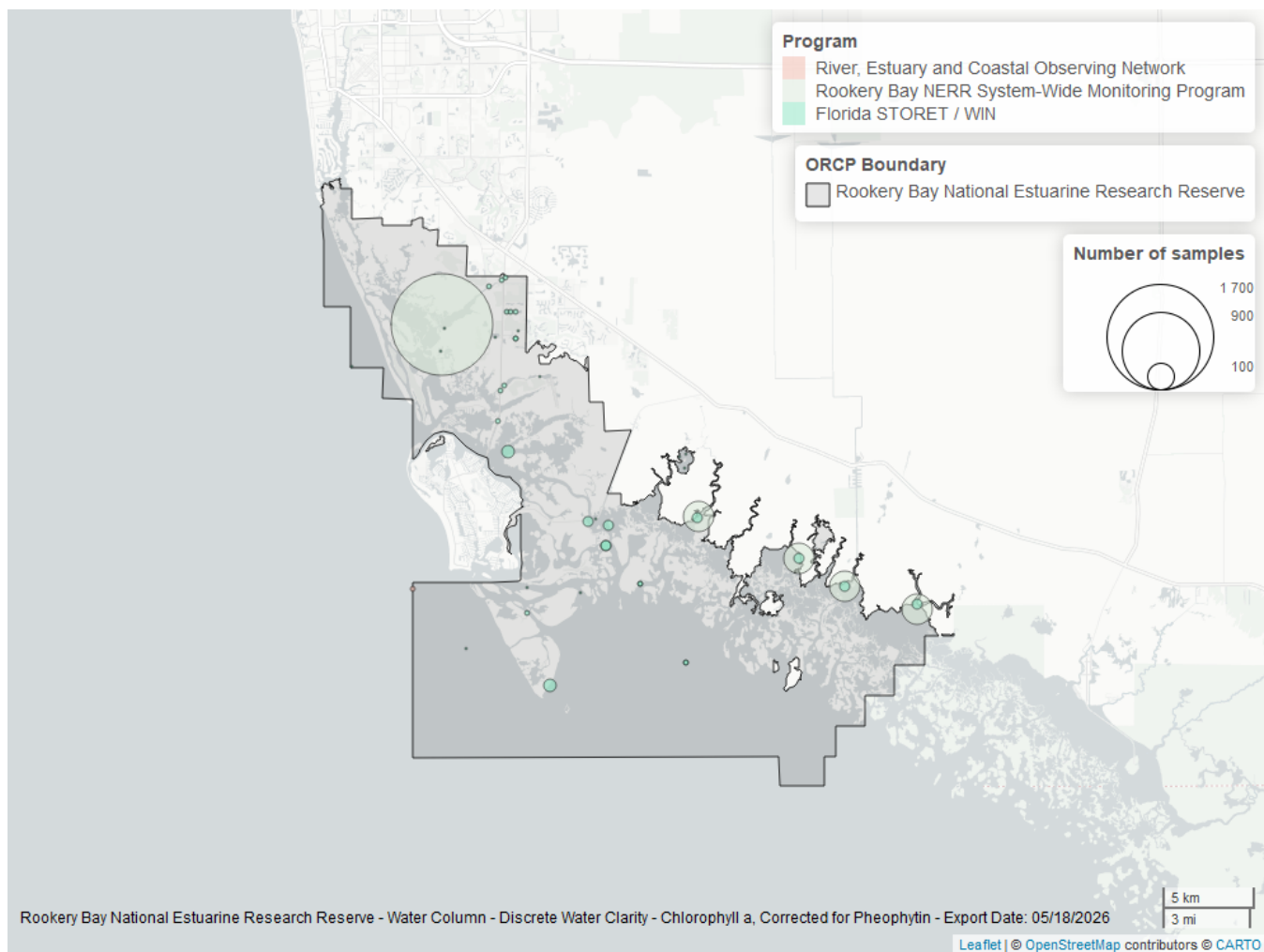


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Chlorophyll a, Corrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
354	2205	2012	2025
5002	226	2002	2025
303	3	2022	2023

Program names:

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

5002 - Florida STORET / WIN²

303 - River, Estuary and Coastal Observing Network³

Chlorophyll a, Uncorrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

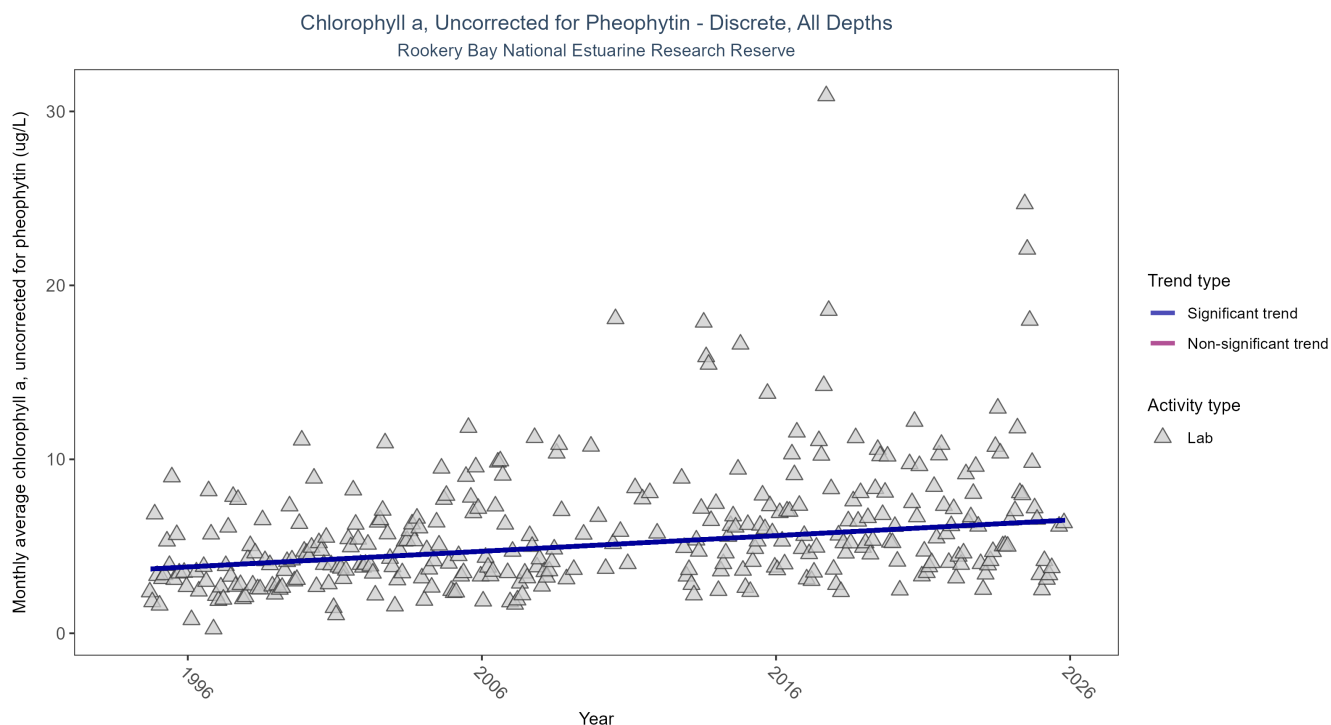


Figure 3: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	5487	32	1994 - 2025	4.4	0.3162	3.6337	0.09	0

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.09 $\mu\text{g/L}$ per year, indicating a decrease in water clarity.

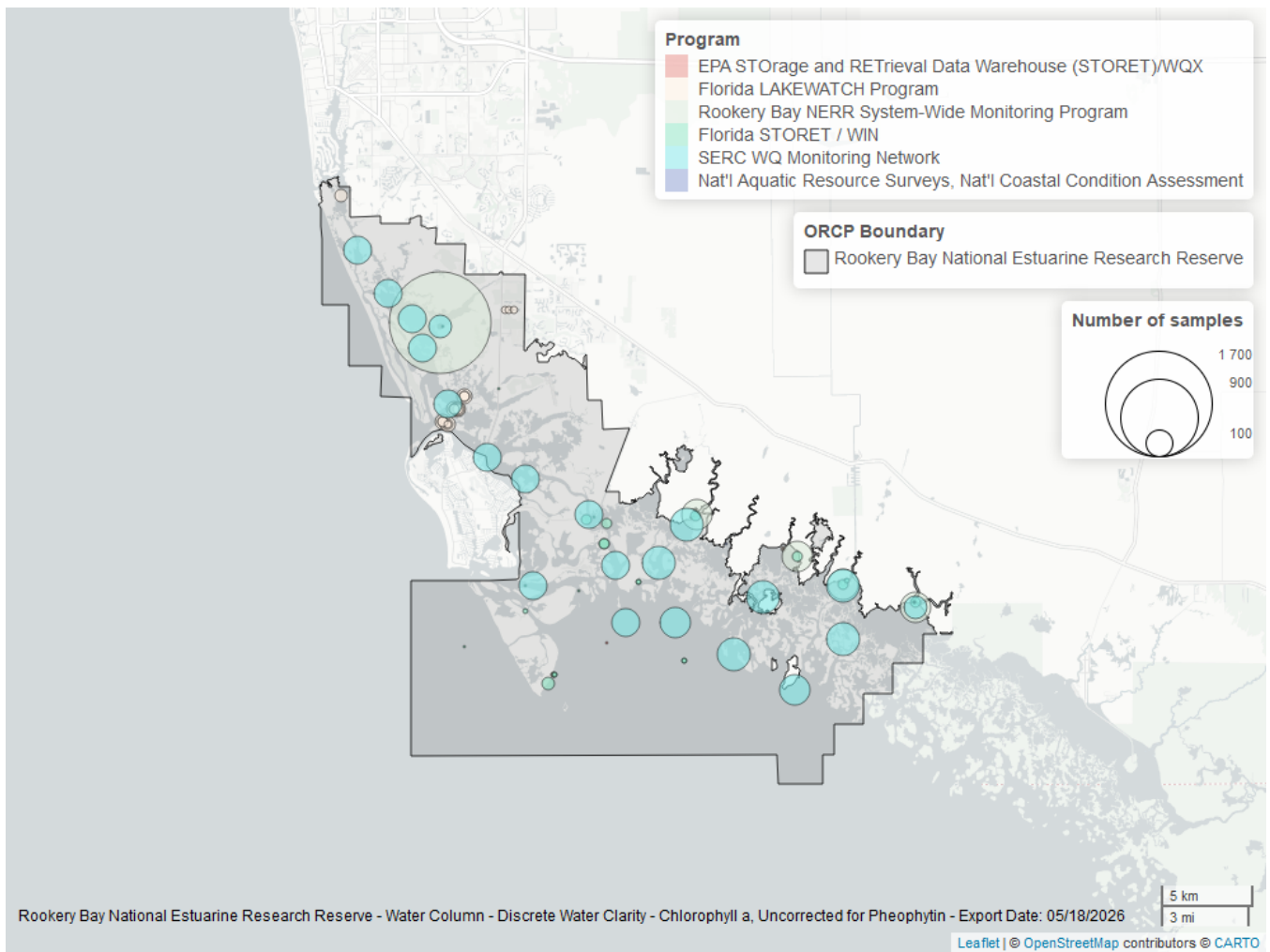


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Chlorophyll a, Uncorrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
509	2746	1994	2008
354	2205	2012	2025
514	413	2001	2017
5002	171	2001	2025
118	1	2010	2010
103	1	2015	2015

Program names:

509 - SERC Water Quality Monitoring Network⁴

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

514 - Florida LAKEWATCH Program⁵

5002 - Florida STORET / WIN²

118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁶

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁷

Colored Dissolved Organic Matter - Discrete

Seasonal Kendall-Tau Trend Analysis

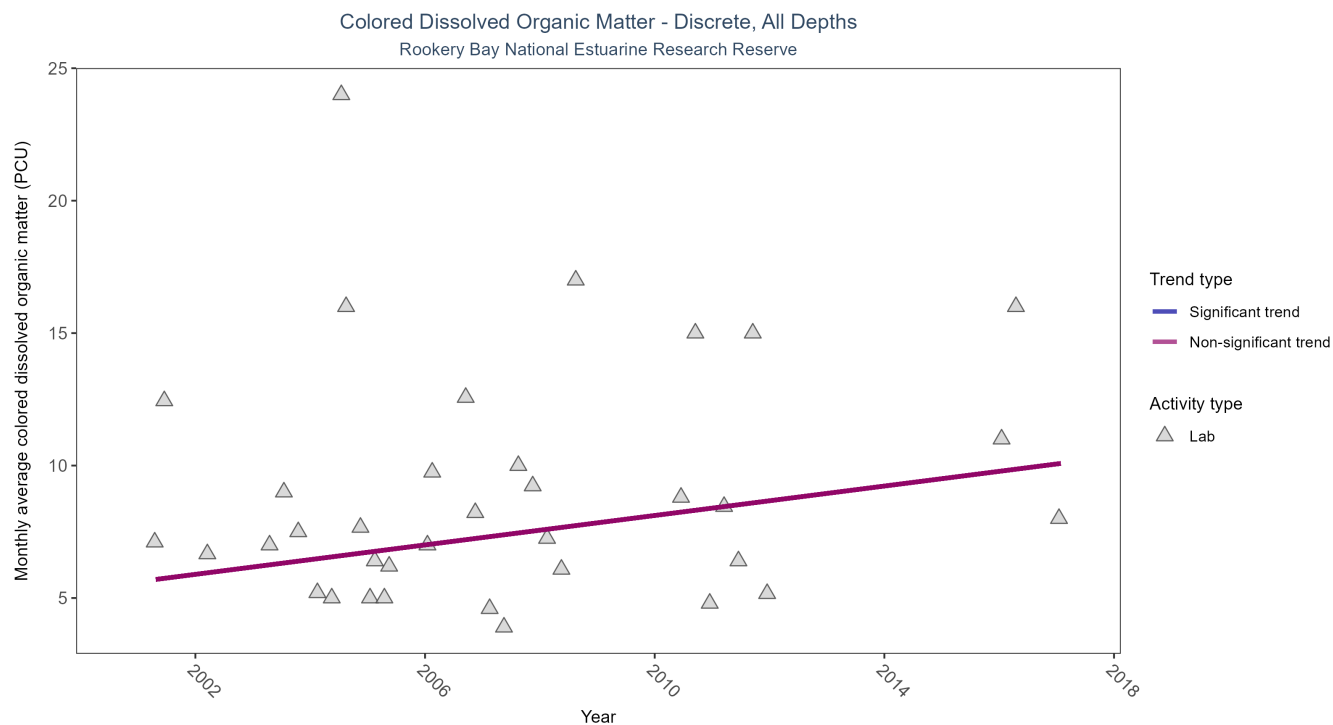


Figure 5: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for Colored Dissolved Organic Matter

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	193	12	2001 - 2017	8	0.3619	5.6154	0.2778	0.1533

Colored dissolved organic matter showed no detectable trend between 2001 and 2017.

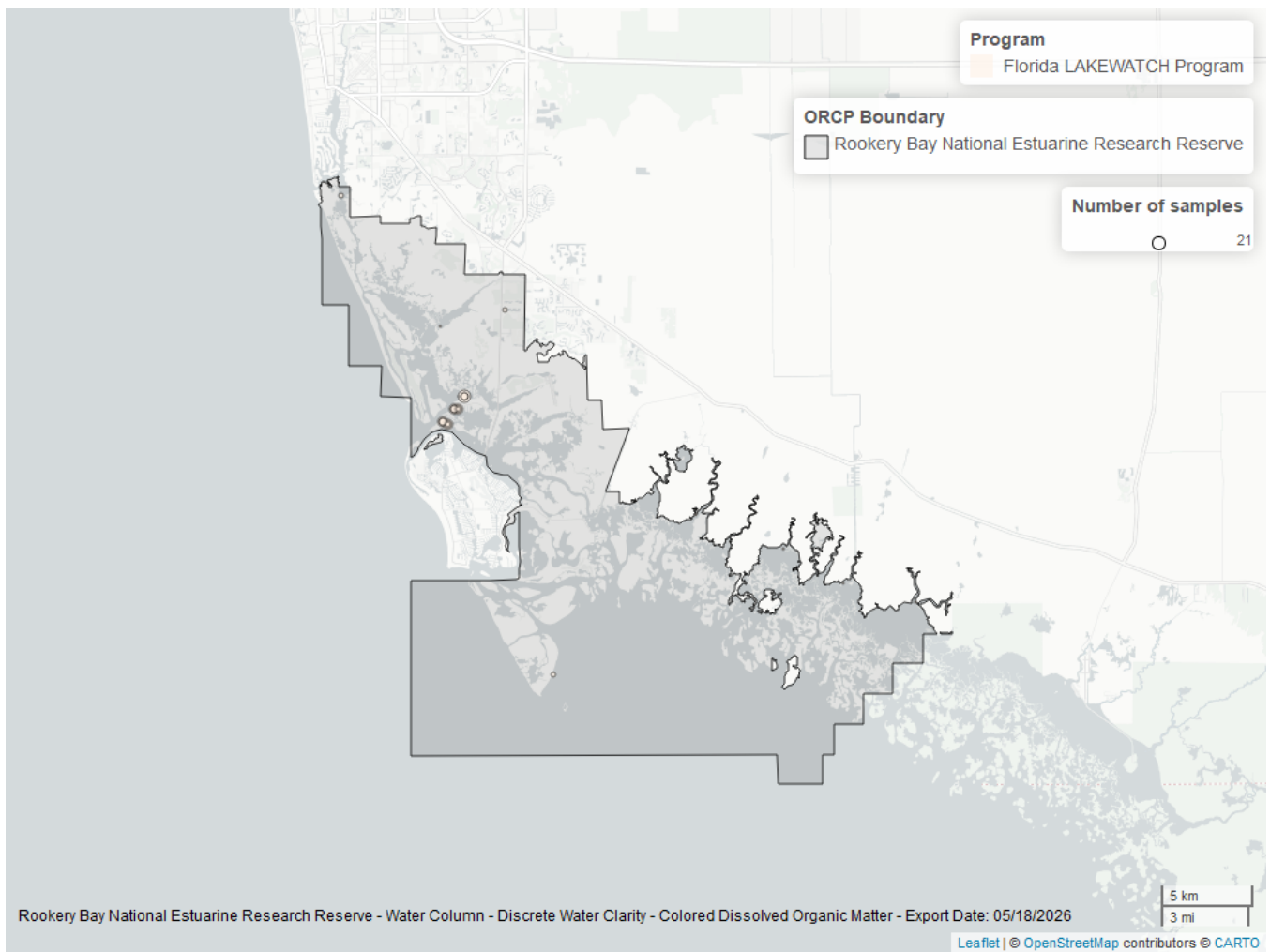


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for Colored Dissolved Organic Matter

ProgramID	N_Data	YearMin	YearMax
514	193	2001	2017

Program names:

514 - Florida LAKEWATCH Program⁵

Dissolved Oxygen - Discrete

Seasonal Kendall-Tau Trend Analysis

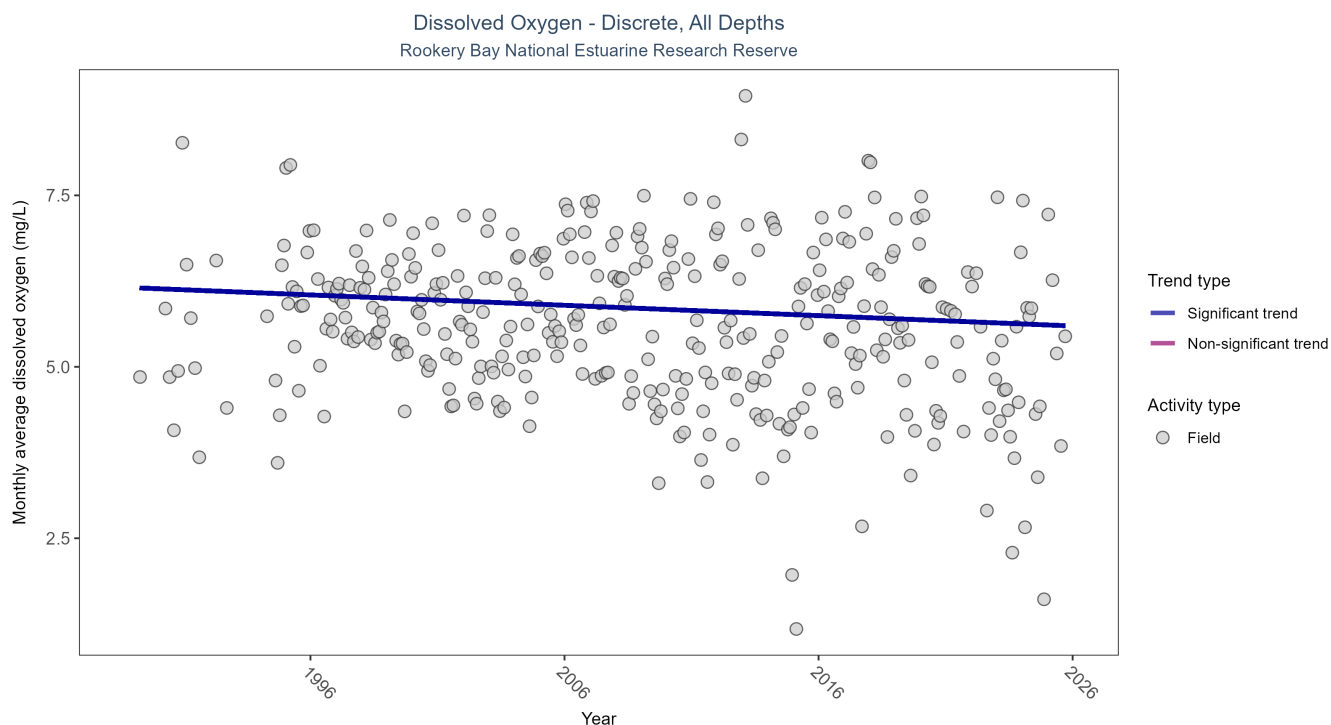


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	14164	36	1989 - 2025	5.8	-0.1175	6.1538	-0.0151	0.0014

Monthly average dissolved oxygen decreased by 0.02 mg/L per year.

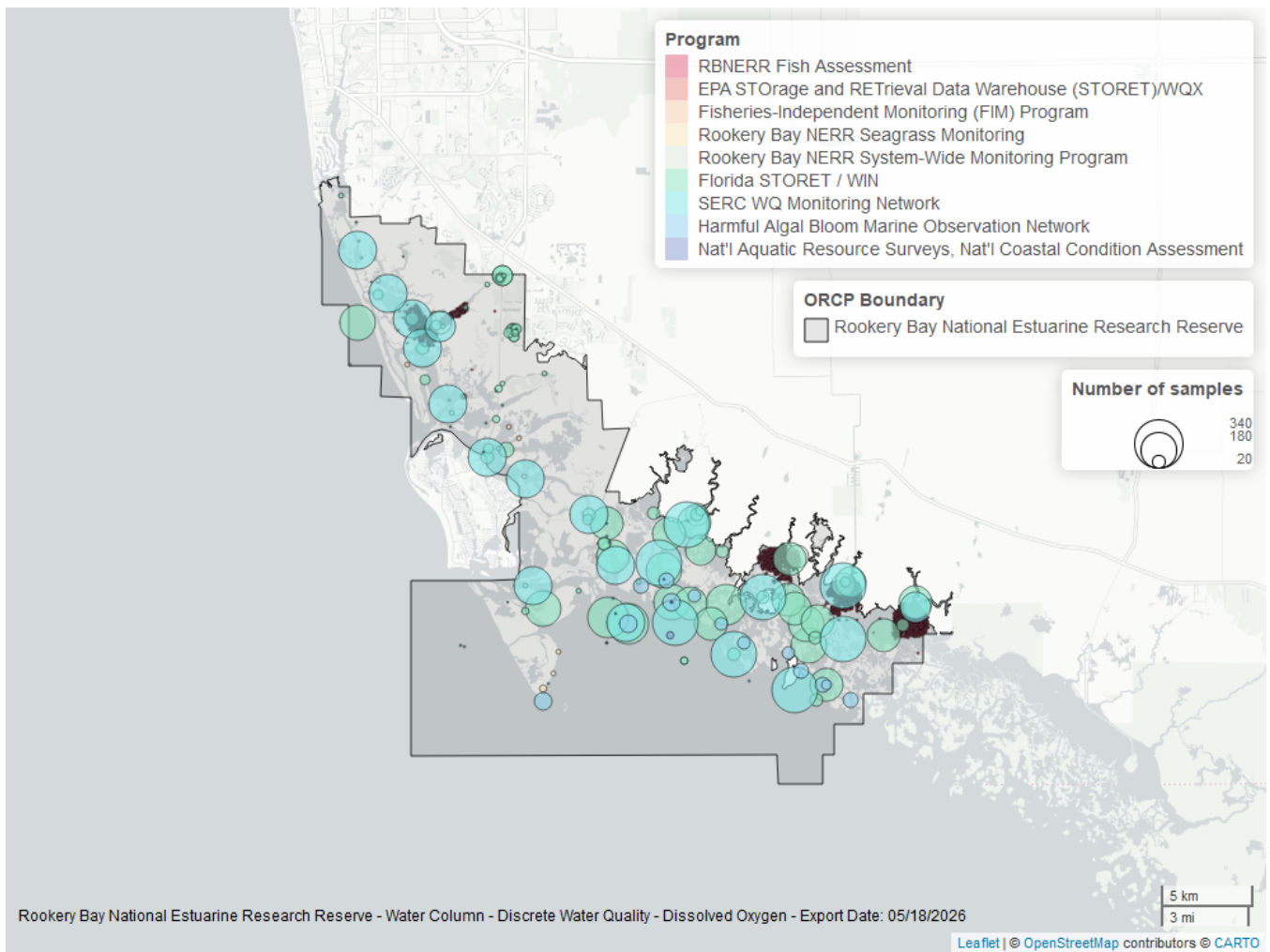


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for Dissolved Oxygen

ProgramID	N_Data	YearMin	YearMax
509	5473	1994	2008
5002	5217	1989	2025
4043	2972	1999	2020
95	442	1997	2018
354	406	2016	2023
572	27	1998	2005
69	22	2001	2001
118	10	2015	2021
103	2	2015	2015

Program names:

509 - SERC Water Quality Monitoring Network⁴

5002 - Florida STORET / WIN²

4043 - RBNERR Fish Assessment⁸

95 - Harmful Algal Bloom Marine Observation Network⁹
354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹
572 - Rookery Bay National Estuarine Research Reserve Seagrass Monitoring¹⁰
69 - Fisheries-Independent Monitoring (FIM) Program¹¹
118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁶
103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁷

Dissolved Oxygen Saturation - Discrete

Seasonal Kendall-Tau Trend Analysis

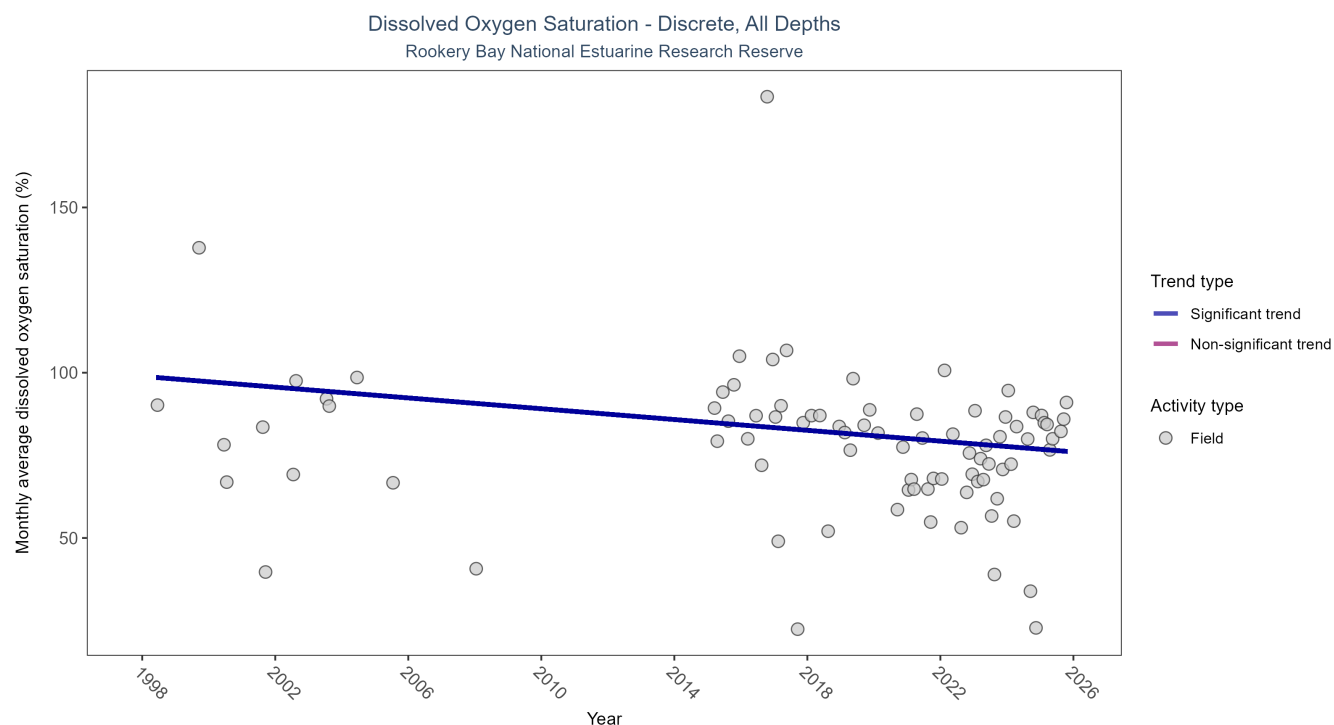


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen Saturation

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	348	20	1998 - 2025	80	-0.2404	98.9033	-0.8167	0.0201

Monthly average dissolved oxygen saturation decreased by 0.82% per year.

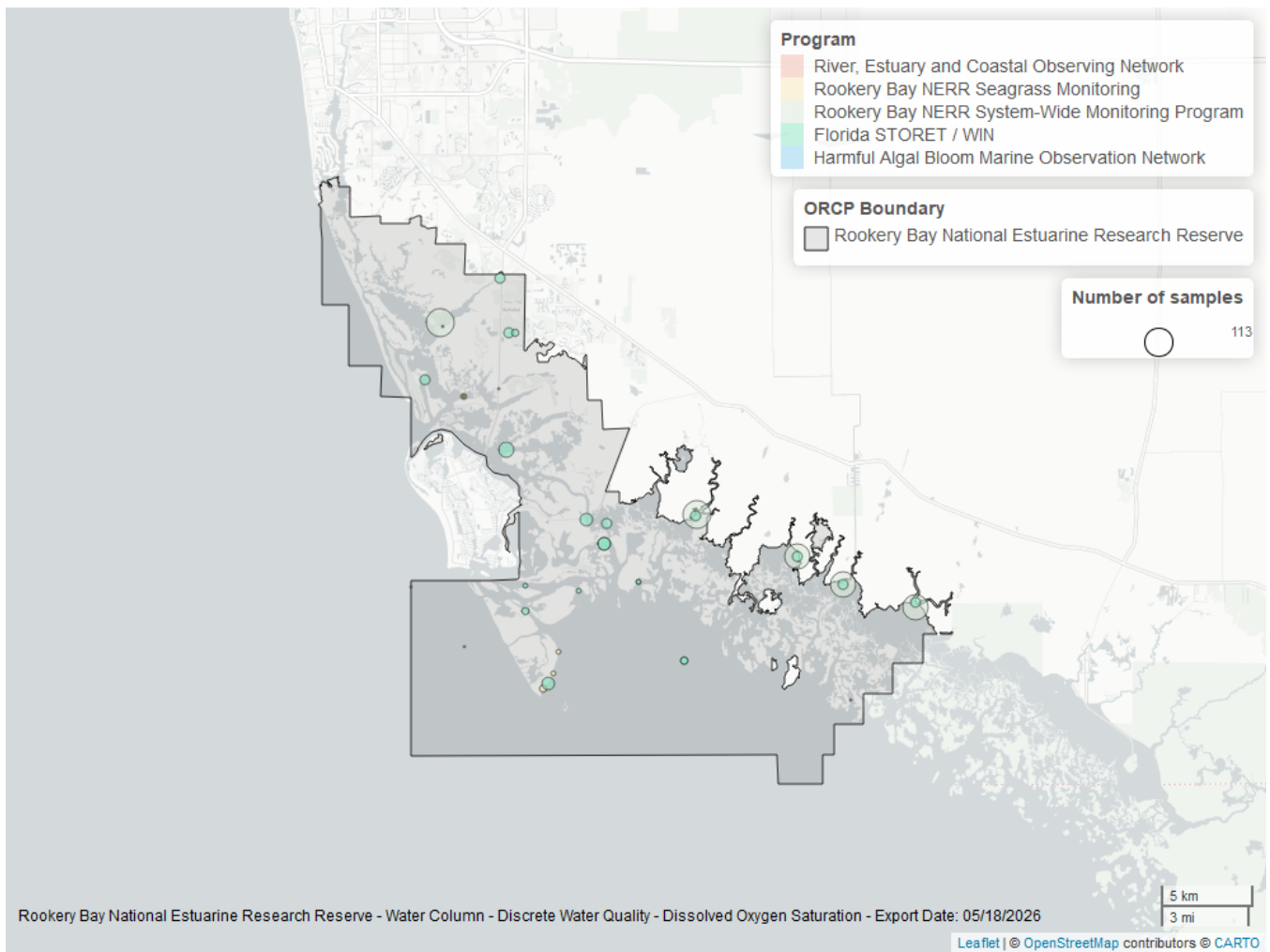


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for Dissolved Oxygen Saturation

ProgramID	N_Data	YearMin	YearMax
354	552	2016	2025
5002	269	2015	2025
572	27	1998	2005
303	1	2023	2023
95	1	2008	2008

Program names:

[354](#) - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

[5002](#) - Florida STORET / WIN²

[572](#) - Rookery Bay National Estuarine Research Reserve Seagrass Monitoring¹⁰

[303](#) - River, Estuary and Coastal Observing Network³

[95](#) - Harmful Algal Bloom Marine Observation Network⁹

pH - Discrete

Seasonal Kendall-Tau Trend Analysis

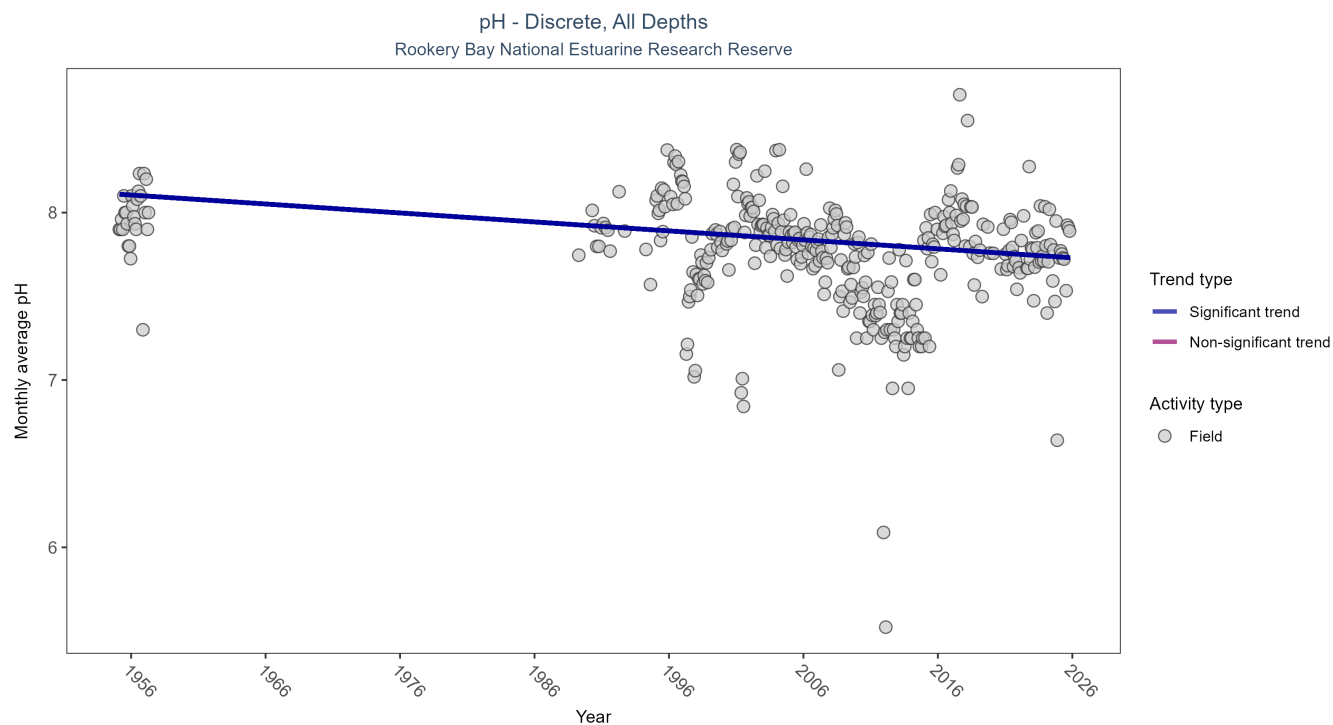


Figure 11: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for pH

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	6488	39	1955 - 2025	7.88	-0.2162	8.1106	-0.0054	0

Monthly average pH decreased by 0.01 pH units per year.

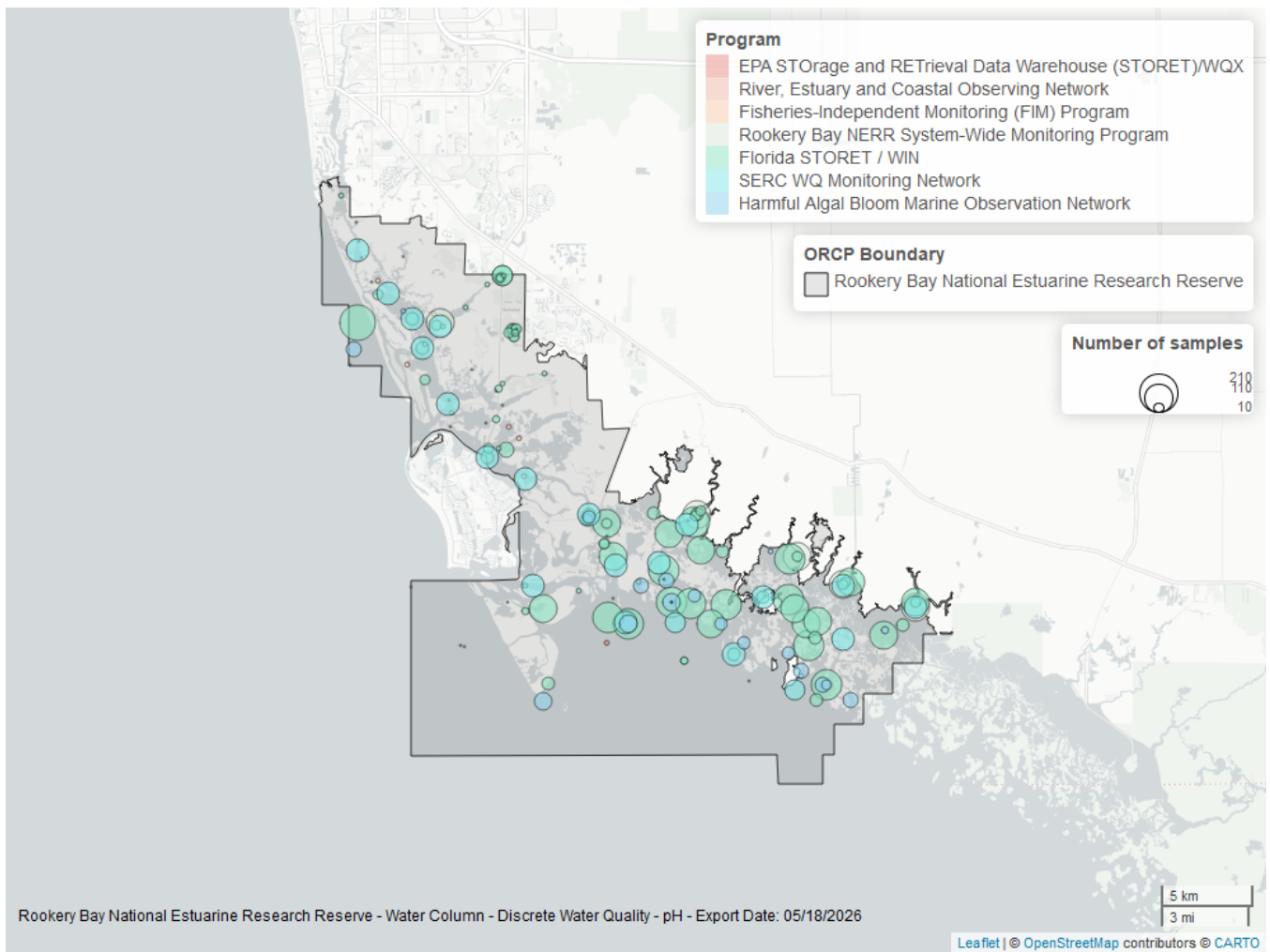


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for pH

ProgramID	N_Data	YearMin	YearMax
5002	4182	1989	2025
509	1718	2001	2008
354	602	2016	2025
95	514	1955	2018
69	22	2001	2001
103	4	2015	2015
303	1	2023	2023

Program names:

5002 - Florida STORET / WIN²

509 - SERC Water Quality Monitoring Network⁴

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

95 - Harmful Algal Bloom Marine Observation Network⁹

69 - Fisheries-Independent Monitoring (FIM) Program¹¹

Salinity - Discrete

Seasonal Kendall-Tau Trend Analysis

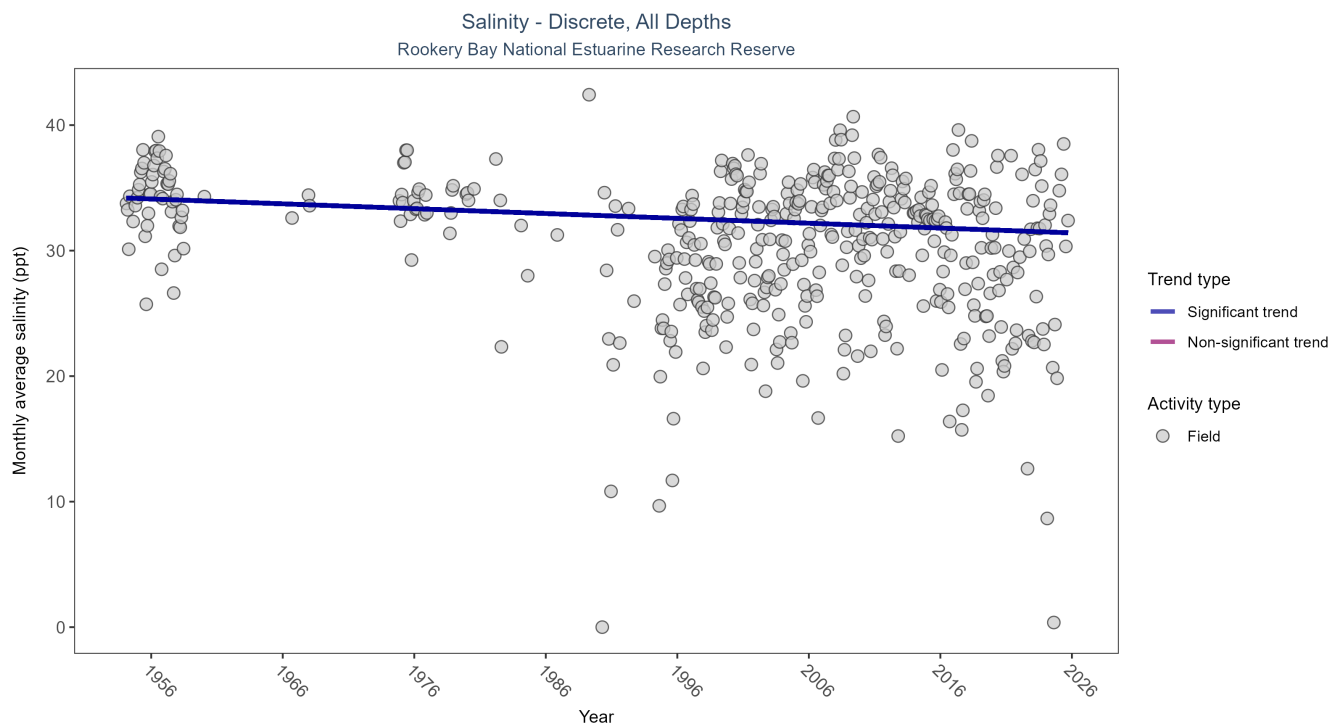


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Salinity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Decreasing trend	15669	54	1954 - 2025	32.8	-0.1391	34.1917	-0.0386	0

Monthly average salinity decreased by 0.04 ppt per year.

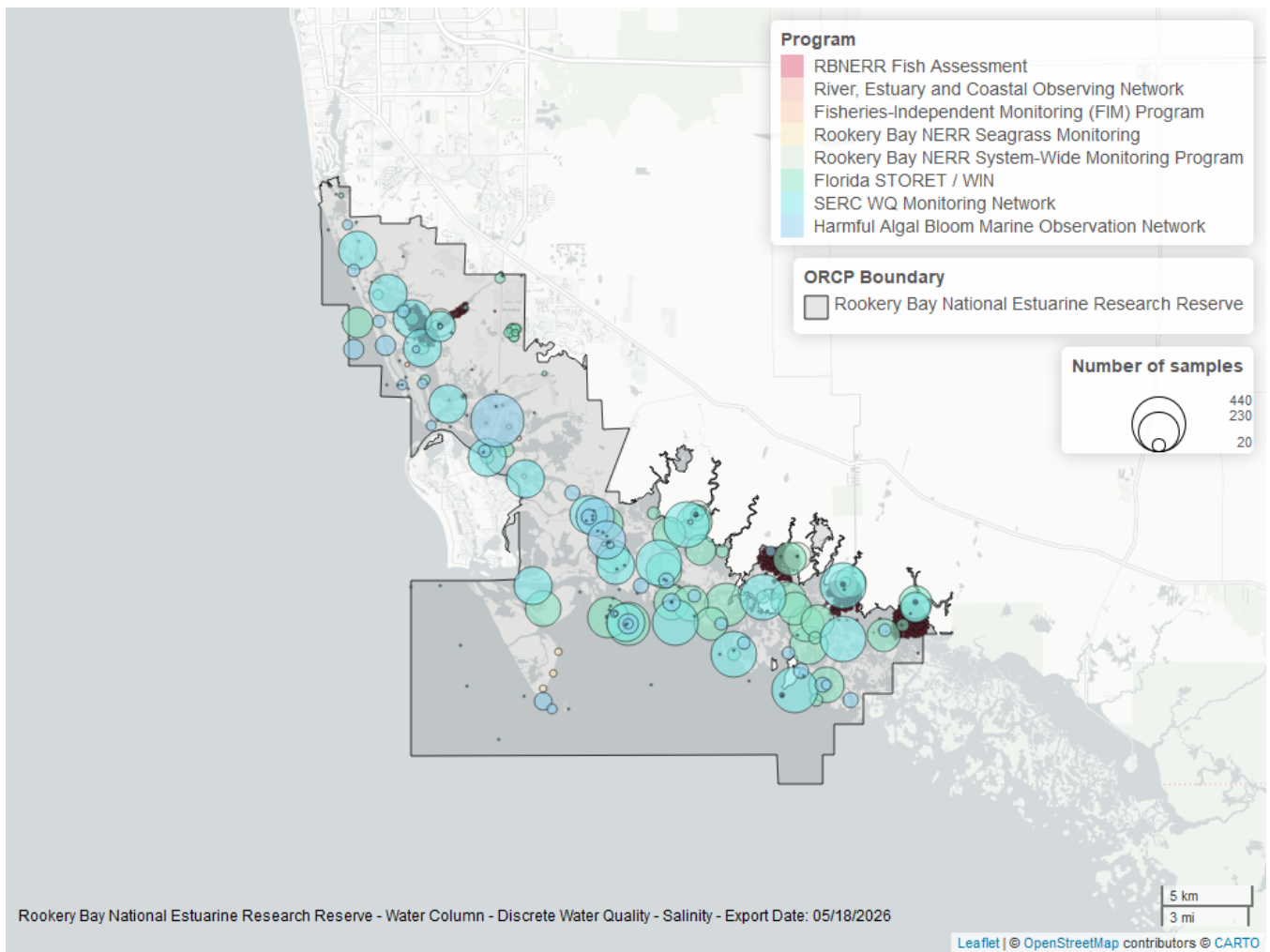


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Salinity

ProgramID	N_Data	YearMin	YearMax
509	5495	1994	2008
5002	5207	1989	2025
4043	3043	1999	2020
95	1890	1954	2018
354	648	2012	2025
572	31	1998	2005
69	22	2001	2001
303	1	2023	2023

Program names:

509 - SERC Water Quality Monitoring Network⁴

5002 - Florida STORET / WIN²

4043 - RBNERR Fish Assessment⁸

95 - Harmful Algal Bloom Marine Observation Network⁹

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

572 - Rookery Bay National Estuarine Research Reserve Seagrass Monitoring¹⁰

69 - Fisheries-Independent Monitoring (FIM) Program¹¹

303 - River, Estuary and Coastal Observing Network³

Total Nitrogen - Discrete

Total Nitrogen Calculation:

The logic for calculated Total Nitrogen was provided by Kevin O'Donnell and colleagues at FDEP (with the help of Jay Silvanima, Watershed Monitoring Section). The following logic is used, in this order, based on the availability of specific nitrogen components.

- 1) $TN = TKN + NO3O2$;
- 2) $TN = TKN + NO3 + NO2$;
- 3) $TN = ORGN + NH4 + NO3O2$;
- 4) $TN = ORGN + NH4 + NO2 + NO3$;
- 5) $TN = TKN + NO3$;
- 6) $TN = ORGN + NH4 + NO3$;

Additional Information:

- Rules for use of sample fraction:
 - Florida Department of Environmental Protection (FDEP) report that if both “Total” and “Dissolved” components are reported, only “Total” is used. If the total is not reported, then the dissolved components are used as a best available replacement.
 - Total nitrogen calculations are done using nitrogen components with the same sample fraction, nitrogen components with mixed total/dissolved sample fractions are not used. In other words, total nitrogen can be calculated when TKN and NO3O2 are both total sample fractions, or when both are dissolved sample fractions. *Future calculations of total nitrogen values may be based on components with mixed sample fractions.*
- Values inserted into data:
 - ParameterName = “Total Nitrogen”
 - SEACAR_QAQCFlagCode = “1Q”
 - SEACAR_QAQC_Description = “SEACAR Calculated”

Seasonal Kendall-Tau Trend Analysis

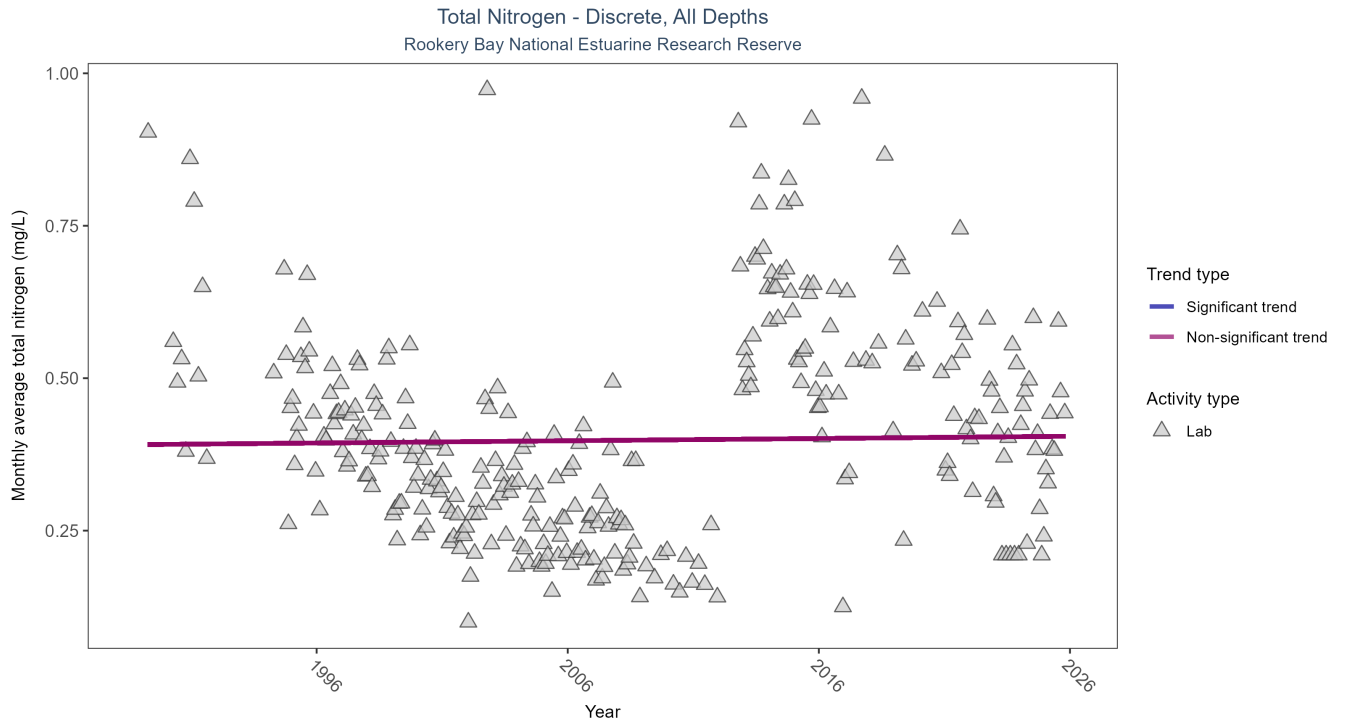


Figure 15: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 20: Seasonal Kendall-Tau Trend Analysis for Total Nitrogen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	4746	35	1989 - 2025	0.3196	0.013	0.3911	0.0004	0.8067

Total nitrogen showed no detectable trend between 1989 and 2025.

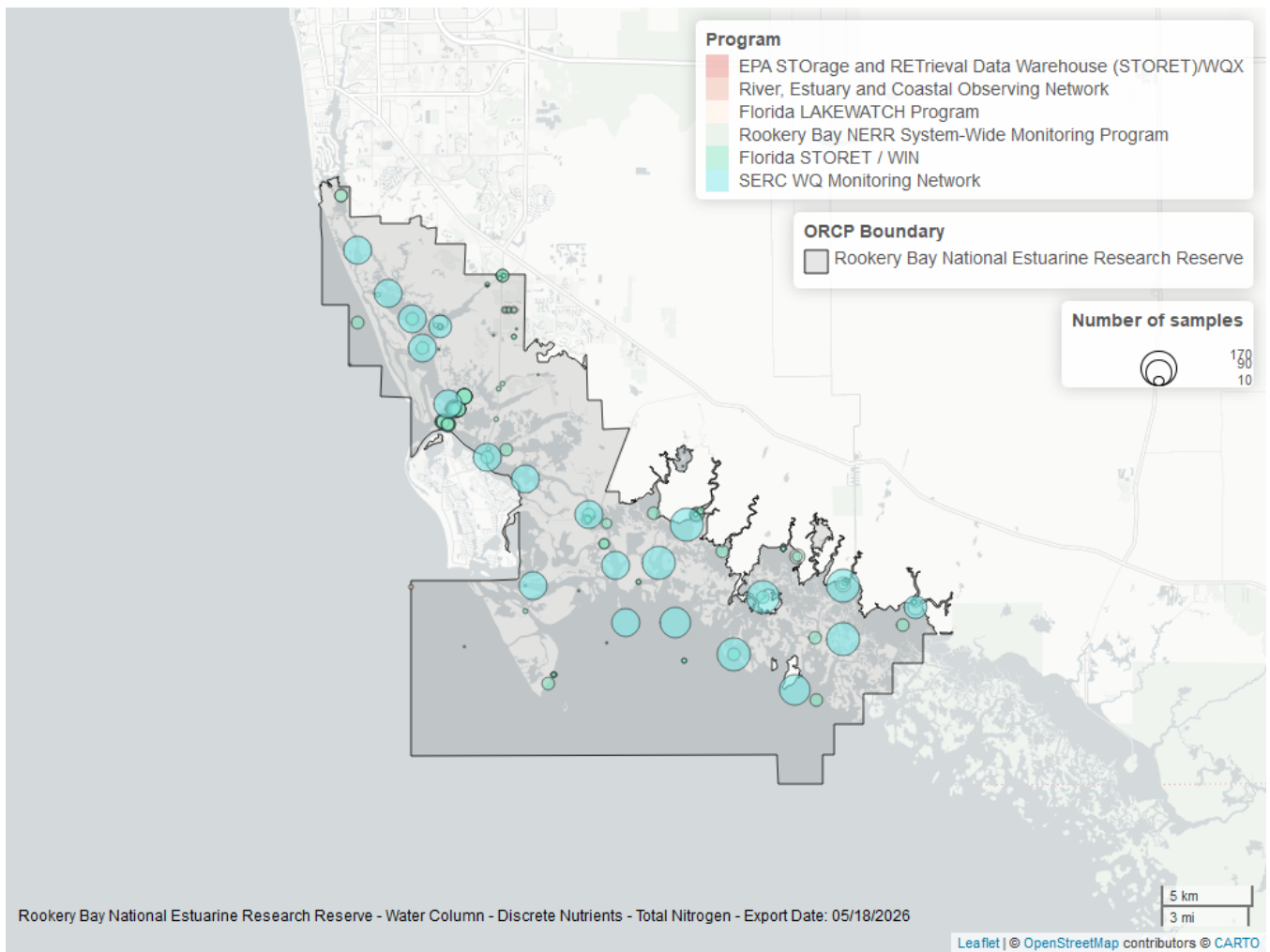


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 21: Programs contributing data for Total Nitrogen

ProgramID	N_Data	YearMin	YearMax
509	2758	1994	2008
5002	1190	1989	2025
514	605	2001	2017
354	197	2012	2017
303	3	2022	2023
103	1	2015	2015

Program names:

509 - SERC Water Quality Monitoring Network⁴

5002 - Florida STORET / WIN²

514 - Florida LAKEWATCH Program⁵

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

303 - River, Estuary and Coastal Observing Network³

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁷

Total Phosphorus - Discrete

Seasonal Kendall-Tau Trend Analysis

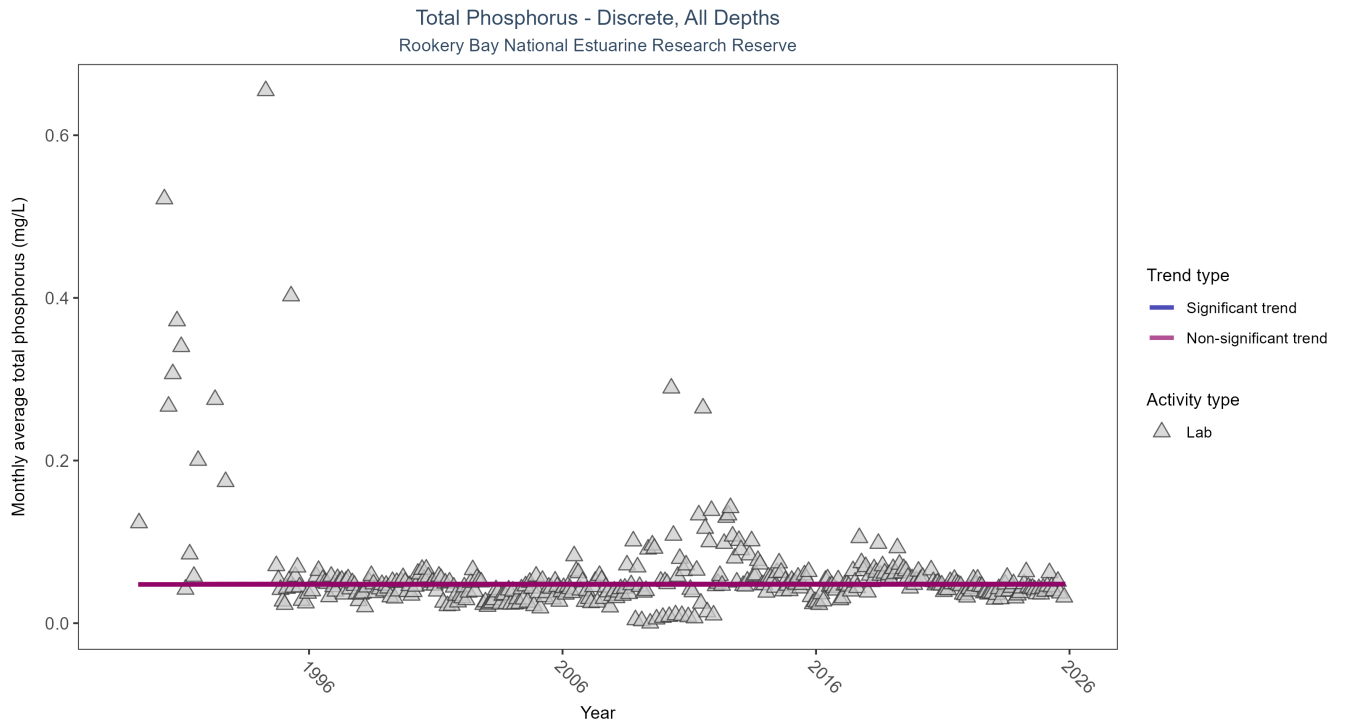


Figure 17: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 22: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	4881	36	1989 - 2025	0.04	0.0093	0.0477	0	0.8776

Total phosphorus showed no detectable trend between 1989 and 2025.

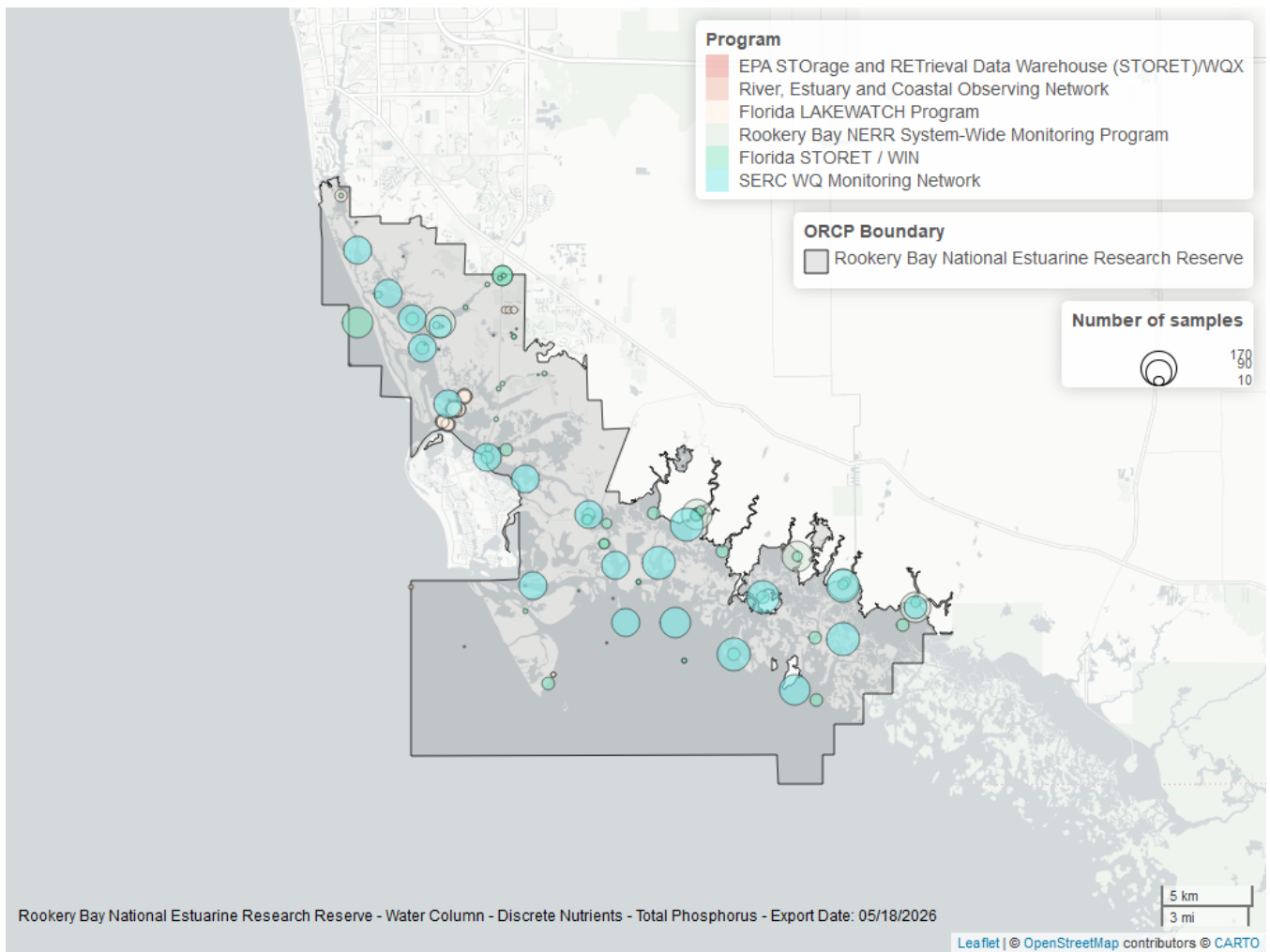


Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 23: Programs contributing data for Total Phosphorus

ProgramID	N_Data	YearMin	YearMax
509	2735	1994	2008
5002	836	1989	2025
354	737	2012	2025
514	597	2001	2017
303	3	2022	2023
103	1	2015	2015

Program names:

509 - SERC Water Quality Monitoring Network⁴

5002 - Florida STORET / WIN²

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

514 - Florida LAKEWATCH Program⁵

303 - River, Estuary and Coastal Observing Network³

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁷

Turbidity - Discrete

Seasonal Kendall-Tau Trend Analysis

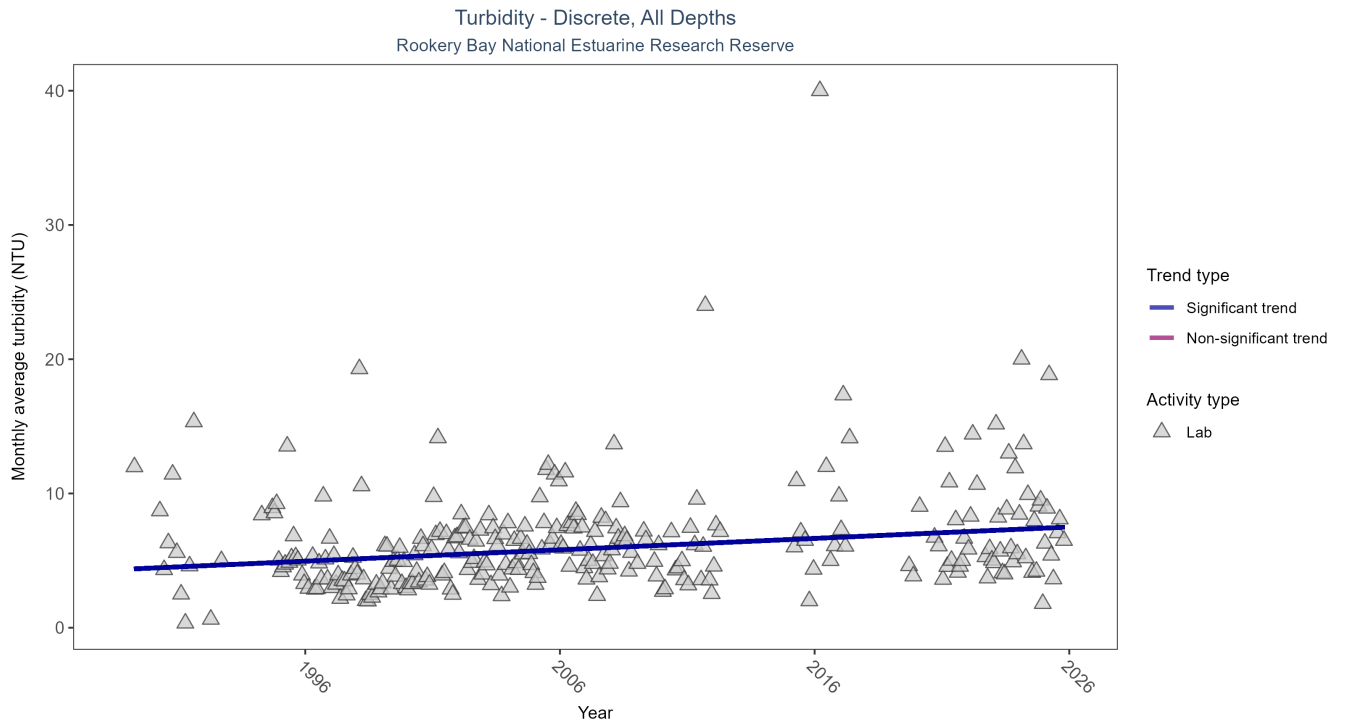


Figure 19: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 24: Seasonal Kendall-Tau Trend Analysis for Turbidity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	6438	33	1989 - 2025	4.3975	0.2068	4.3599	0.0849	0

Monthly average turbidity increased by 0.08 NTU per year, indicating a decrease in water clarity.

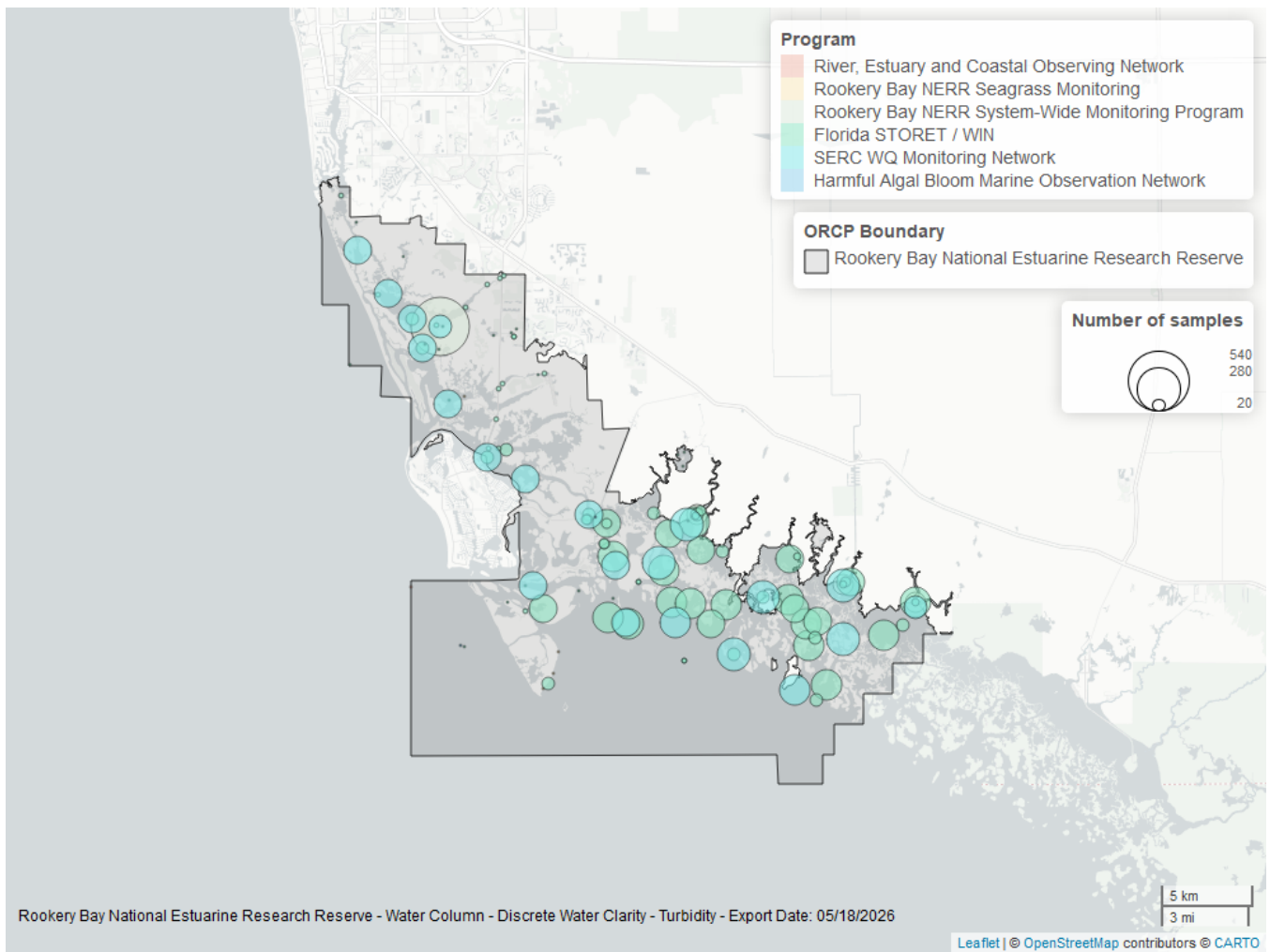


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 25: Programs contributing data for Turbidity

ProgramID	N_Data	YearMin	YearMax
5002	3707	1989	2025
509	2746	1994	2008
354	734	2002	2006
572	4	2000	2003
95	3	2003	2011
303	1	2023	2023

Program names:

5002 - Florida STORET / WIN²

509 - SERC Water Quality Monitoring Network⁴

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

572 - Rookery Bay National Estuarine Research Reserve Seagrass Monitoring¹⁰

95 - Harmful Algal Bloom Marine Observation Network⁹

303 - River, Estuary and Coastal Observing Network³

Water Temperature - Discrete

Seasonal Kendall-Tau Trend Analysis

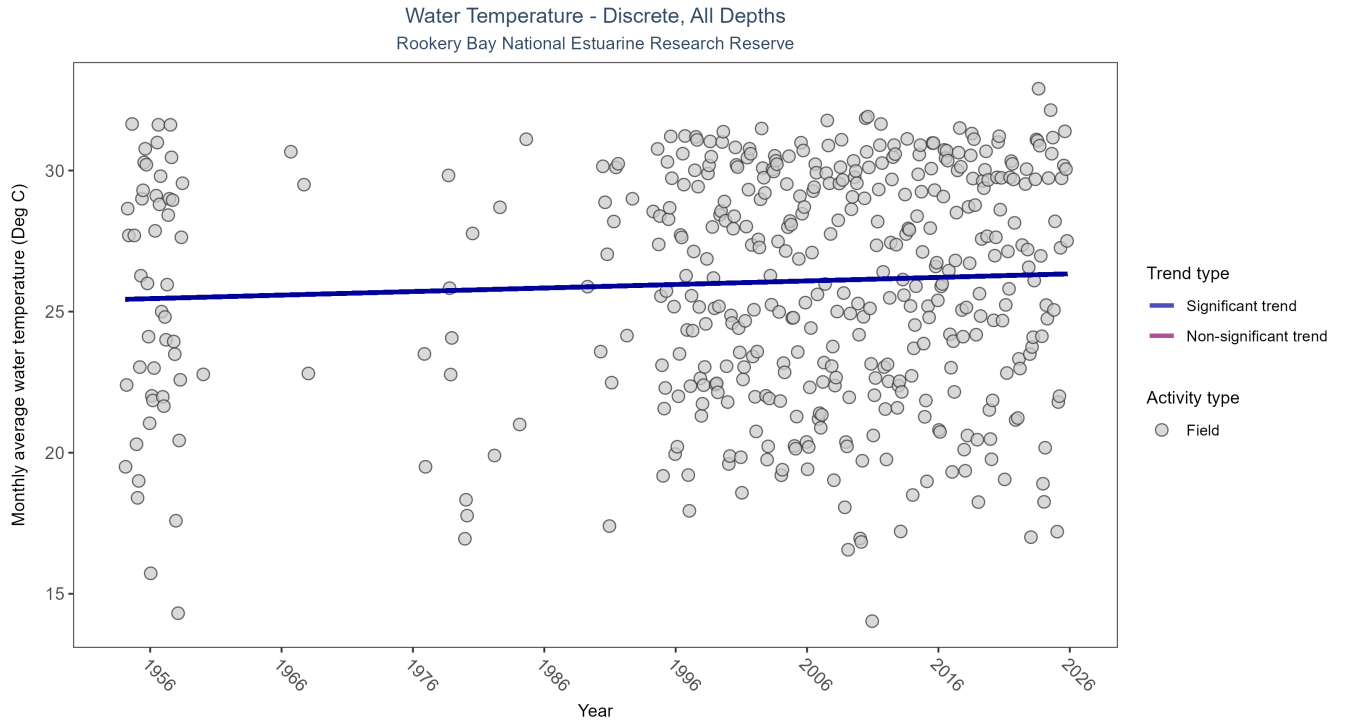


Figure 21: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 26: Seasonal Kendall-Tau Trend Analysis for Water Temperature

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	15682	51	1954 - 2025	26.63	0.1066	25.4348	0.0126	0.0012

Monthly average water temperature increased by 0.01°C per year.

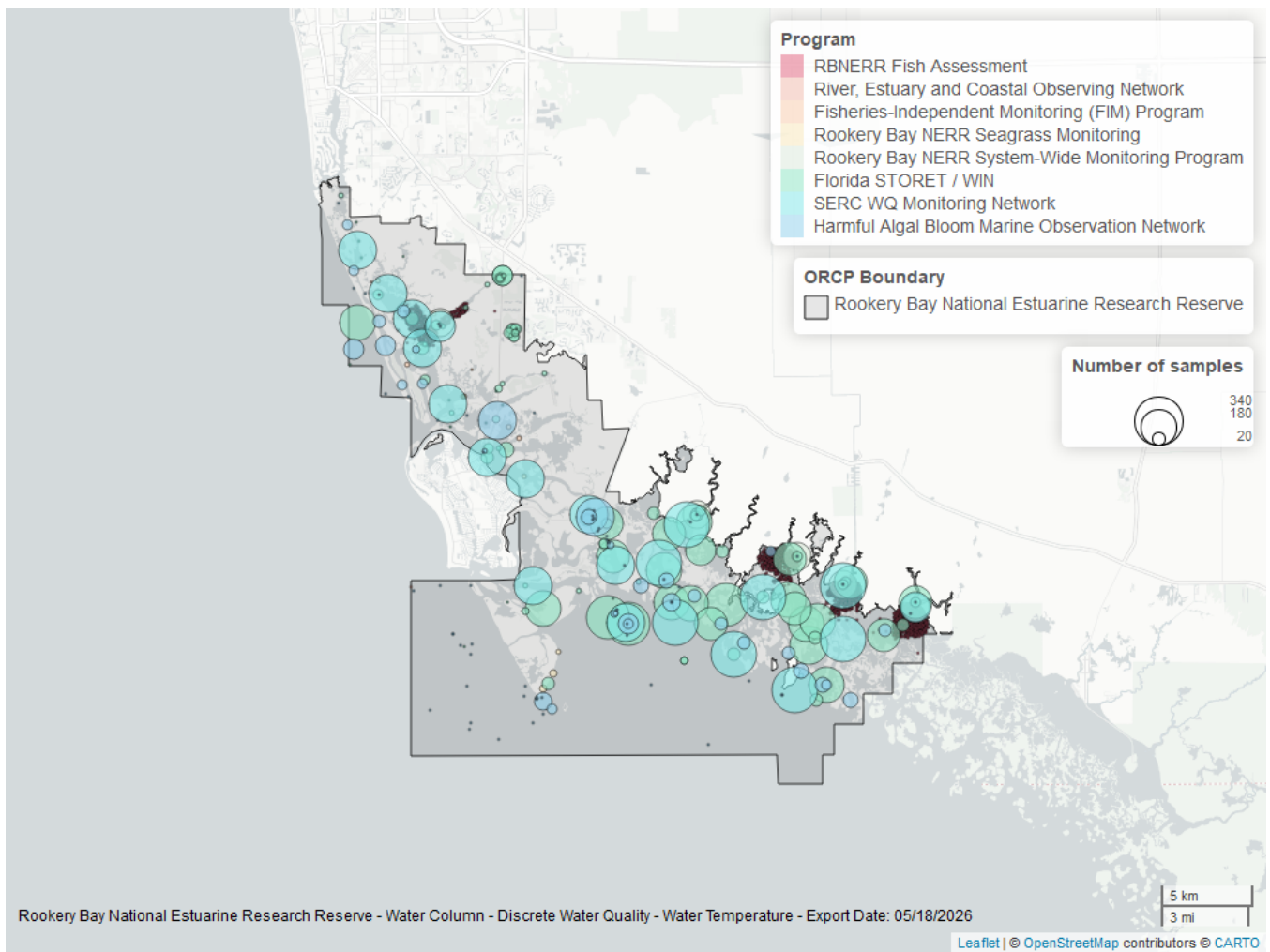


Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 27: Programs contributing data for Water Temperature

ProgramID	N_Data	YearMin	YearMax
5002	5710	1989	2025
509	5498	1994	2008
4043	3036	1999	2020
95	1336	1954	2018
354	615	2012	2025
572	30	1998	2005
69	22	2001	2001
303	1	2023	2023

Program names:

5002 - Florida STORET / WIN²

509 - SERC Water Quality Monitoring Network⁴

4043 - RBNERR Fish Assessment⁸

95 - Harmful Algal Bloom Marine Observation Network⁹

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹
572 - Rookery Bay National Estuarine Research Reserve Seagrass Monitoring¹⁰
69 - Fisheries-Independent Monitoring (FIM) Program¹¹
303 - River, Estuary and Coastal Observing Network³

Water Quality - Continuous

The following files were used in the continuous analysis:

- *Combined_WQ_WC_NUT_cont_Chlorophyll_a_Uncorrected_for_Pheophytin-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen_Saturation-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Fluorescent_Dissolved_Organic_Matter-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_pH-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Salinity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Specific_Conductivity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Turbidity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Water_Temperature-2026-Mar-06.txt*

Continuous monitoring locations in Rookery Bay National Estuarine Research Reserve

Table 28: Station overview for Continuous parameters by Program

<i>ProgramID</i>	<i>ProgramLocationID</i>	<i>Years of Data</i>	<i>Use in Analysis</i>	<i>Parameters</i>
7	02291330	4	FALSE	Water Temperature, Salinity
7	255123081321300	8	TRUE	Specific Conductivity
7	255123081321300	11	TRUE	Water Temperature, Salinity
7	255138081321701	10	TRUE	Water Temperature , Salinity , Specific Conductivity
7	255432081303900	17	TRUE	Specific Conductivity
7	255432081303900	21	TRUE	Water Temperature, Salinity
7	255443081314700	5	FALSE	Water Temperature, Salinity
7	255532081314300	3	FALSE	Water Temperature, Salinity
7	255534081324000	17	TRUE	Specific Conductivity
7	255534081324000	20	TRUE	Water Temperature, Salinity
7	255654081350200	17	TRUE	Specific Conductivity
7	255654081350200	20	TRUE	Water Temperature, Salinity
7	255732081363700	5	TRUE	Water Temperature, Salinity
354	rkbbmwq	2	FALSE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
354	rkbfwbq	25	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
354	rkbfwbq	25	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
354	rkblhwq	26	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
354	rkblmbwq	27	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
354	rkbpwbq	11	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
354	rkbulhwq	5	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity

Program names:

7 - National Water Information System¹²

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

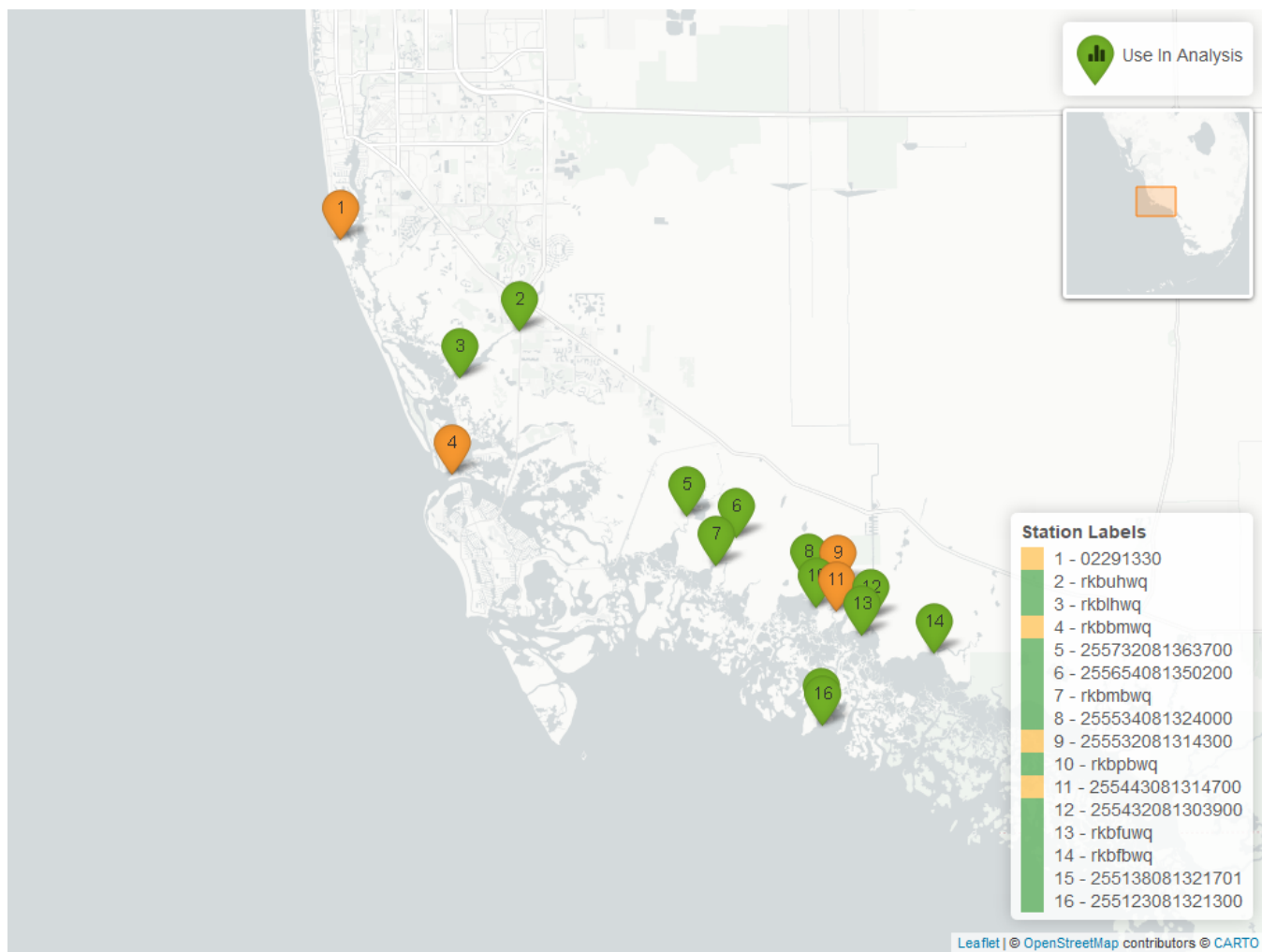


Figure 23: Map showing continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. Sites marked as *Use In Analysis* (green) are featured in this report.

pH - Continuous

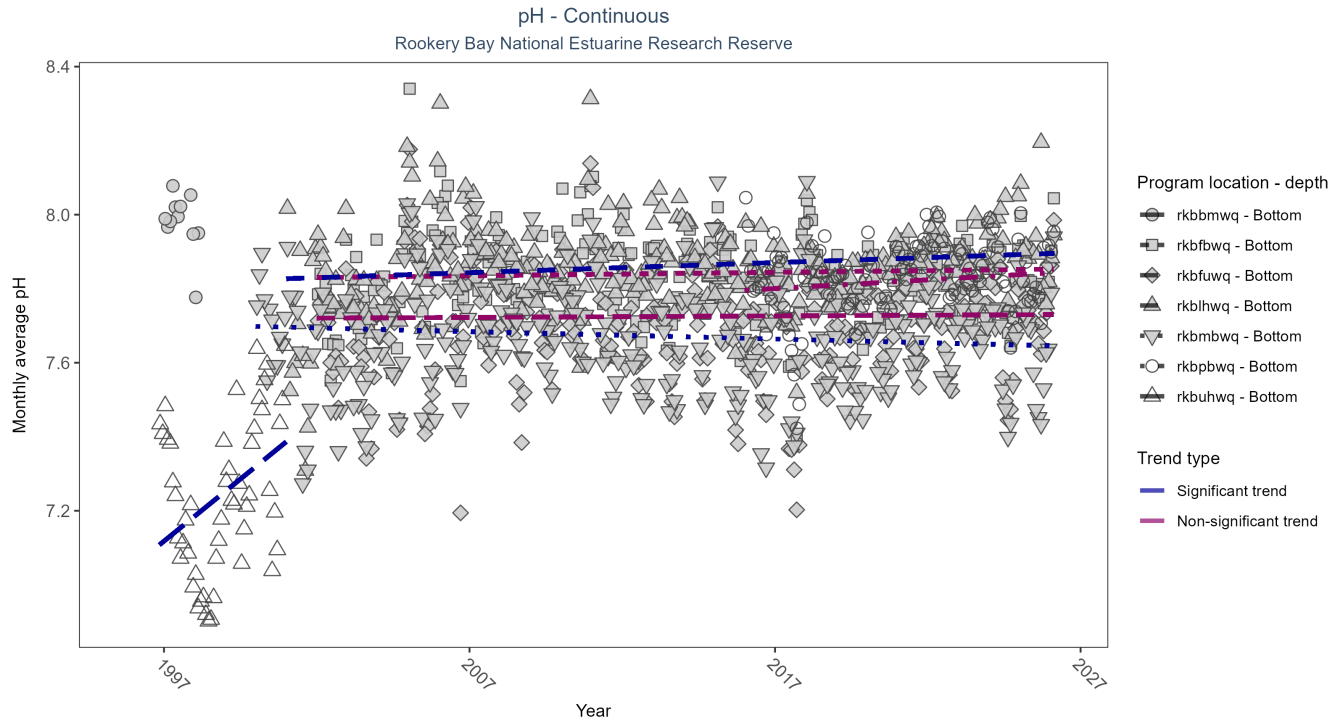


Figure 24: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 29: Seasonal Kendall-Tau Results for pH - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfwq	No significant trend	670684	25	2002 - 2026	7.8	0.05	7.83	0	0.21
rkbfuwq	No significant trend	689115	25	2002 - 2026	7.7	0.02	7.72	0	0.61
rkbbmwq	Significantly decreasing trend	727383	27	2000 - 2026	7.7	-0.1	7.7	0	0.02
rkbpbwq	No significant trend	328113	11	2016 - 2026	7.8	0.13	7.8	0	0.08
rkbbmwq	Insufficient data to calculate trend	12610	2	1997 - 1998	8.0	-	-	-	-
rkblhwq	Significantly increasing trend	672354	26	2001 - 2026	7.9	0.15	7.83	0	0
rkbuwq	Significantly increasing trend	65814	5	1996 - 2000	7.2	0.37	7.05	0.07	0.01

At two program locations, monthly average pH increased by less than 0.01 pH units per year at one site and by 0.07 pH units per year at the other. At one program location, monthly average pH decreased by less than 0.01 pH units per year. No detectable change in monthly average pH was observed at three locations. There was insufficient data to fit a model for one location.

Dissolved Oxygen Saturation - Continuous

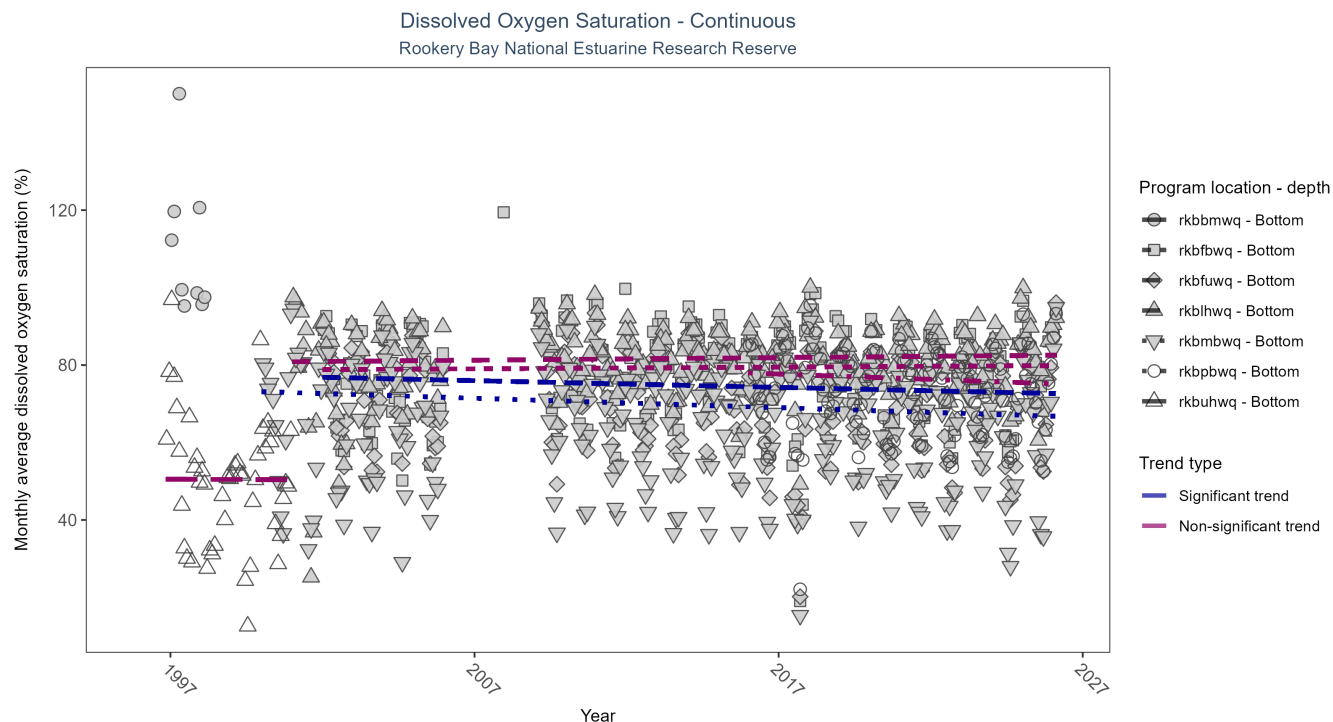


Figure 25: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 30: Seasonal Kendall-Tau Results for Dissolved Oxygen Saturation - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkfbwq	No significant trend	608215	23	2002 - 2026	79.0	0.04	78.81	0.04	0.43
rkbfuwq	Significantly decreasing trend	636774	22	2002 - 2026	72.6	-0.2	76.84	-0.17	0
rkbbmwq	Significantly decreasing trend	663443	24	2000 - 2026	65.3	-0.21	73.13	-0.24	0
rkbpbwq	No significant trend	336249	11	2016 - 2026	72.7	-0.13	77.96	-0.28	0.08
rkbbmwq	Insufficient data to calculate trend	10441	2	1997 - 1998	102.4	-	-	-	-
rkblhwq	No significant trend	625534	23	2001 - 2026	82.1	0.07	80.86	0.07	0.13
rkbuhwq	No significant trend	58164	5	1996 - 2000	49.7	0	50.53	-0.02	1

At two program locations, monthly average dissolved oxygen saturation decreased by 0.17% per year at one site and by 0.24% per year at the other. No detectable change in monthly average dissolved oxygen saturation was observed at four locations. There was insufficient data to fit a model for one location.

Dissolved Oxygen - Continuous

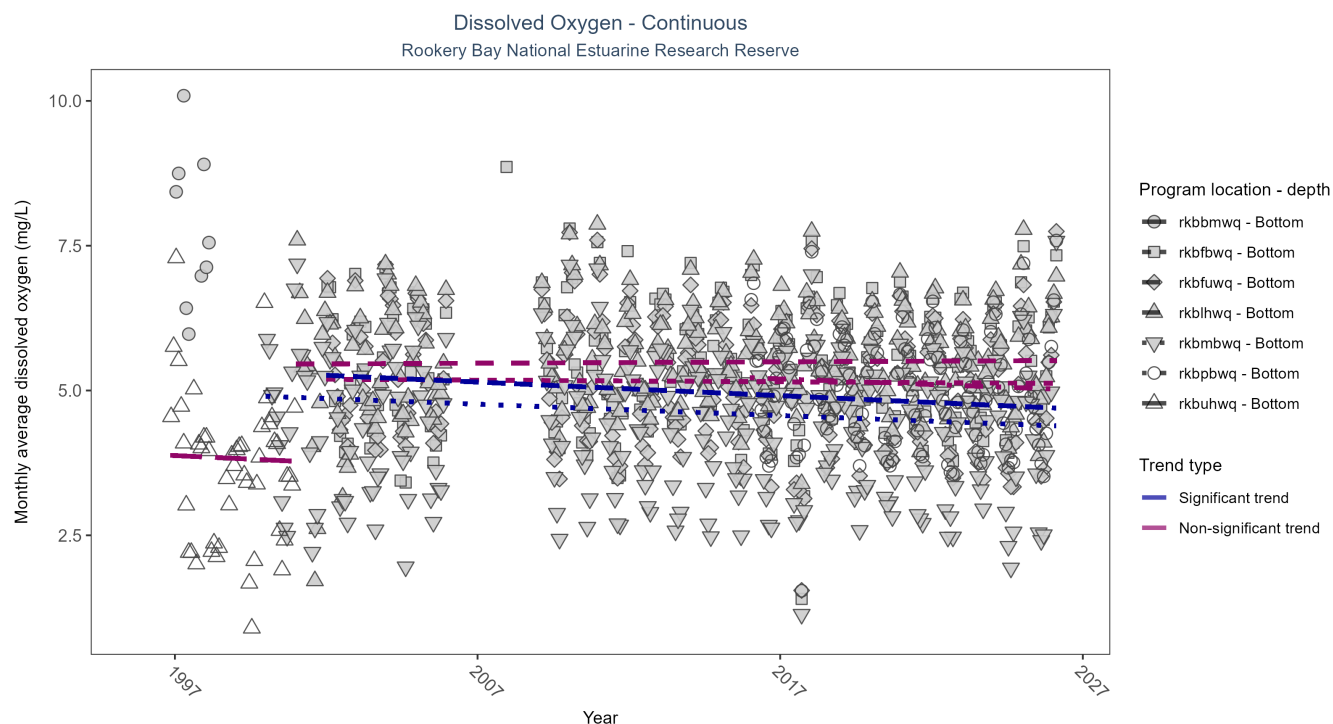


Figure 26: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 31: Seasonal Kendall-Tau Results for Dissolved Oxygen - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfwq	No significant trend	603909	23	2002 - 2026	5.4	-0.02	5.19	0	0.62
rkbfuwq	Significantly decreasing trend	636515	22	2002 - 2026	5.1	-0.29	5.26	-0.02	0
rkbbmwq	Significantly decreasing trend	657119	24	2000 - 2026	4.4	-0.24	4.9	-0.02	0
rkbpbwq	No significant trend	332812	11	2016 - 2026	4.9	-0.14	5.22	-0.02	0.08
rkbbmwq	Insufficient data to calculate trend	10441	2	1997 - 1998	7.2	-	-	-	-
rkblhwq	No significant trend	613216	23	2001 - 2026	5.6	0.03	5.46	0	0.52
rkbuwq	No significant trend	58164	5	1996 - 2000	3.6	0.01	3.9	-0.02	1

At two program locations, monthly average dissolved oxygen decreased by 0.02 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at four locations. There was insufficient data to fit a model for one location.

Turbidity - Continuous

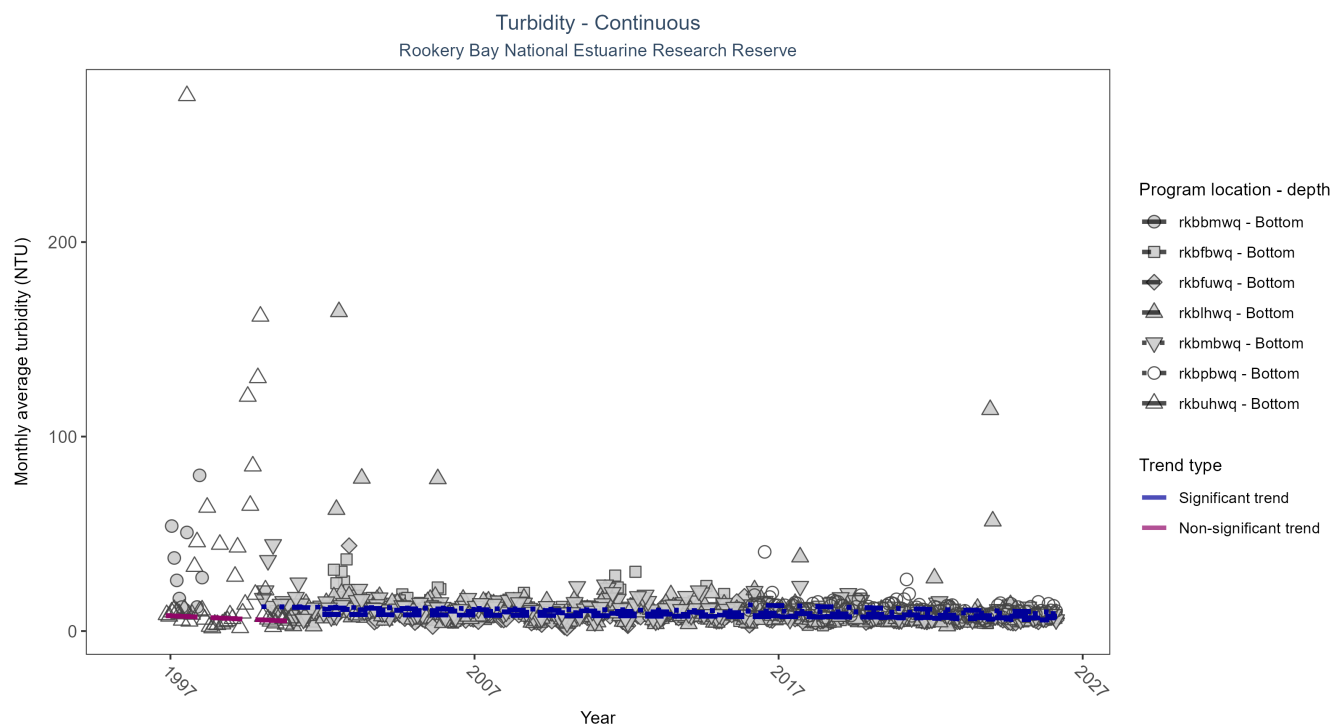


Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 32: Seasonal Kendall-Tau Results for Turbidity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfbwq	Significantly decreasing trend	682915	25	2002 - 2026	7	-0.4	11.3	-0.23	0
rkbfbwq	Significantly decreasing trend	657778	25	2002 - 2026	6	-0.19	8.57	-0.08	0
rkbmbwq	Significantly decreasing trend	714237	27	2000 - 2026	9	-0.24	12.56	-0.14	0
rkbpbwq	Significantly decreasing trend	331299	11	2016 - 2026	9	-0.31	13.47	-0.37	0
rkbmbwq	Insufficient data to calculate trend	10654	2	1997 - 1998	11	-	-	-	-
rkbhlwq	Significantly decreasing trend	647527	26	2001 - 2026	8	-0.28	12.22	-0.19	0
rkbuhwq	No significant trend	61608	5	1996 - 2000	5	-0.11	8.57	-0.7	0.52

At five program locations, monthly average turbidity decreased between 0.08 and 0.37 NTU per year. No detectable change in monthly average turbidity was observed at one location. There was insufficient data to fit a model for one location.

Water Temperature - Continuous - Program 7

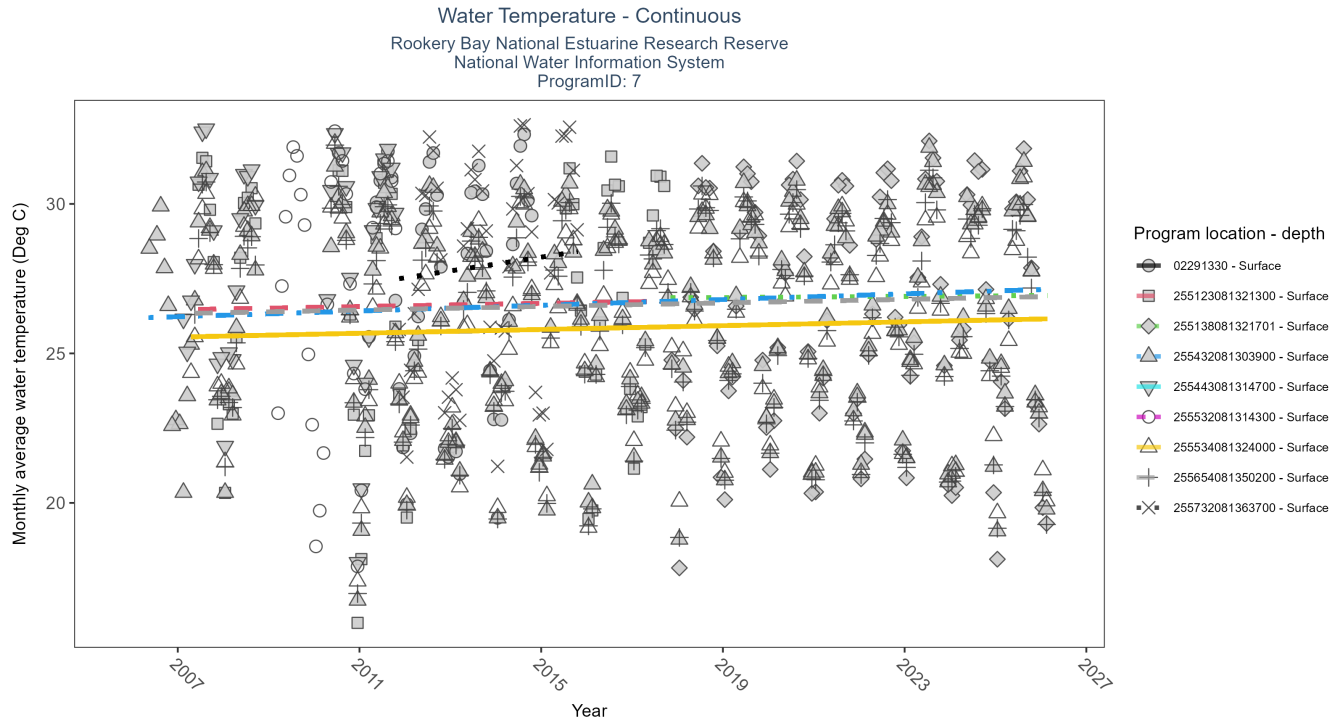


Figure 28: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 33: Seasonal Kendall-Tau Results for Water Temperature - Program 7

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
255123081321300	No significant trend	1818	8	2007 - 2017	27.3	0.1	26.46	0.03	0.38
255138081321701	No significant trend	3005	10	2017 - 2026	27.0	0.02	26.85	0.01	0.92
255432081303900	Significantly increasing trend	6527	20	2006 - 2026	27.0	0.17	26.18	0.05	0
255443081314700	Insufficient data to calculate trend	2011	4	2007 - 2011	29.3	-	-	-	-
255532081314300	Insufficient data to calculate trend	906	3	2009 - 2011	29.3	-	-	-	-
255534081324000	Significantly increasing trend	6225	19	2007 - 2026	26.6	0.14	25.55	0.03	0.01
255654081350200	Significantly increasing trend	6222	19	2007 - 2026	26.7	0.11	26.33	0.03	0.03
255732081363700	No significant trend	1435	5	2011 - 2015	28.4	0.24	27.3	0.23	0.11
02291330	Insufficient data to calculate trend	1901	4	2011 - 2014	28.3	-	-	-	-

At seven program locations, monthly average water temperature increased between 0.03 and 0.06°C per year. No detectable change in monthly average water temperature was observed at five locations. There was insufficient data to fit a model for four locations.

Water Temperature - Continuous - Program 354

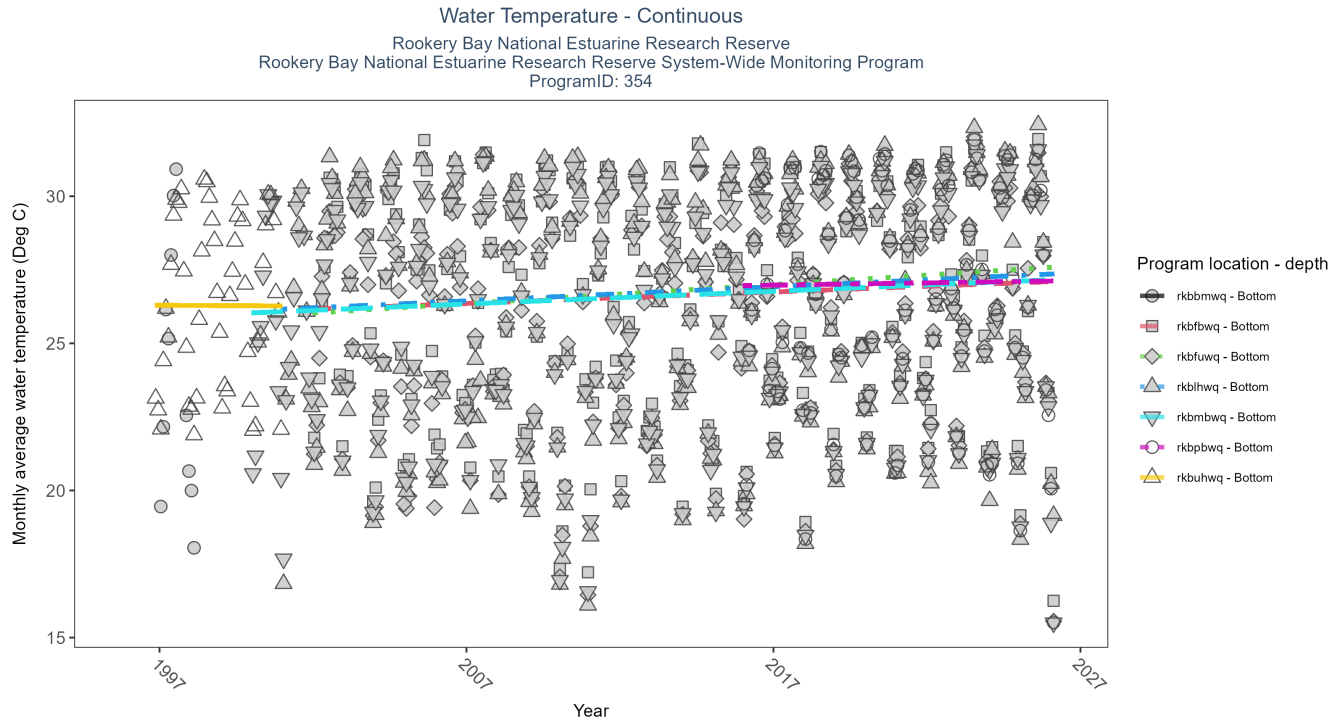


Figure 29: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 34: Seasonal Kendall-Tau Results for Water Temperature - Program 354

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfwq	Significantly increasing trend	719021	25	2002 - 2026	27.0	0.22	26.16	0.04	0
rkbfuwq	Significantly increasing trend	727501	25	2002 - 2026	26.9	0.26	26.03	0.06	0
rkmbbwq	Significantly increasing trend	762033	27	2000 - 2026	26.8	0.25	26.04	0.04	0
rkbpbwq	No significant trend	337833	11	2016 - 2026	27.4	0.05	26.96	0.02	0.55
rkbbmwq	Insufficient data to calculate trend	12610	2	1997 - 1998	23.8	-	-	-	-
rkblhwq	Significantly increasing trend	731519	26	2001 - 2026	26.9	0.24	26.16	0.05	0
rkbuhwq	No significant trend	68971	5	1996 - 2000	26.8	-0.01	26.3	-0.01	1

At seven program locations, monthly average water temperature increased between 0.03 and 0.06°C per year. No detectable change in monthly average water temperature was observed at five locations. There was insufficient data to fit a model for four locations.

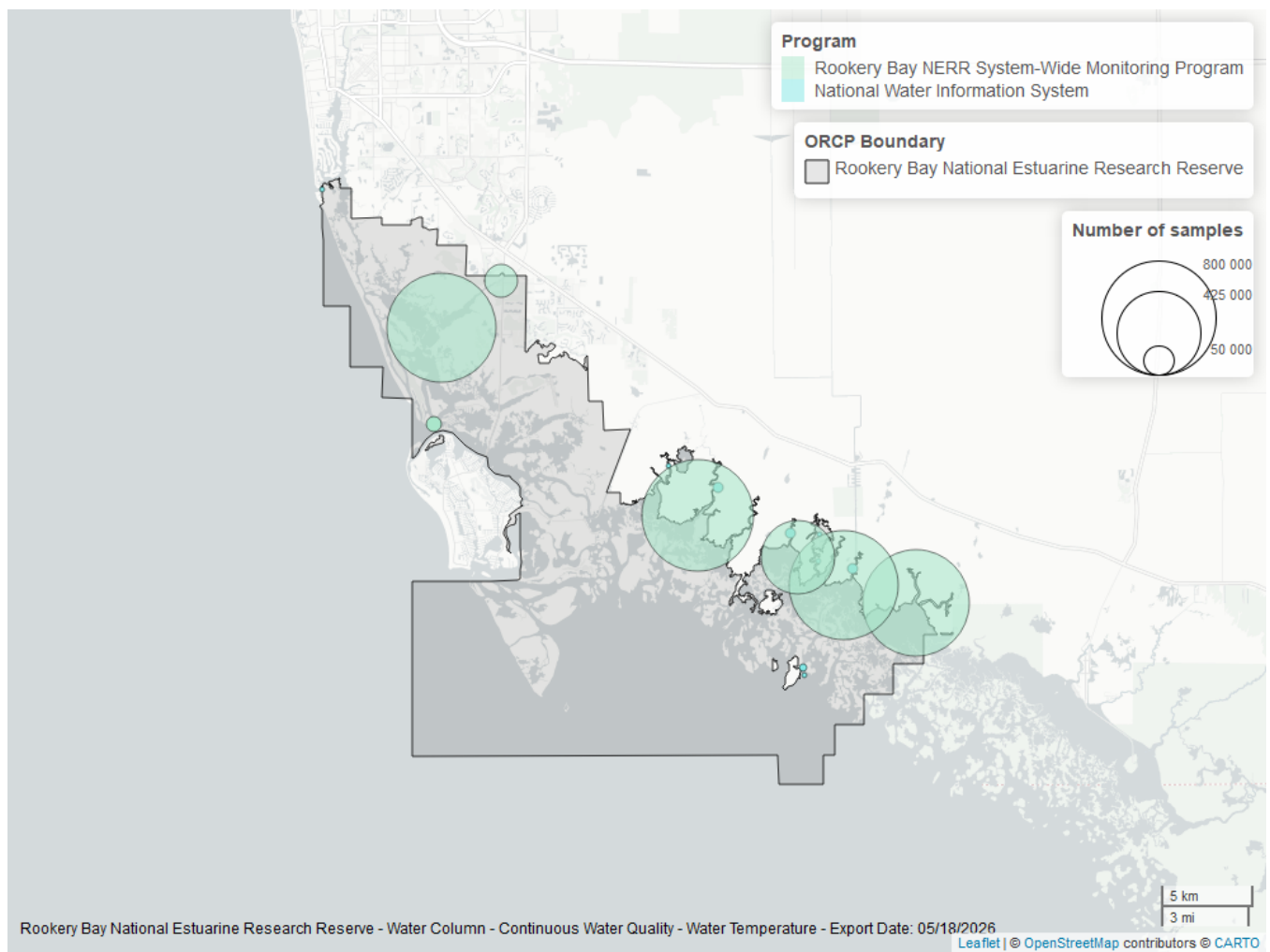


Figure 30: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous - Program 7

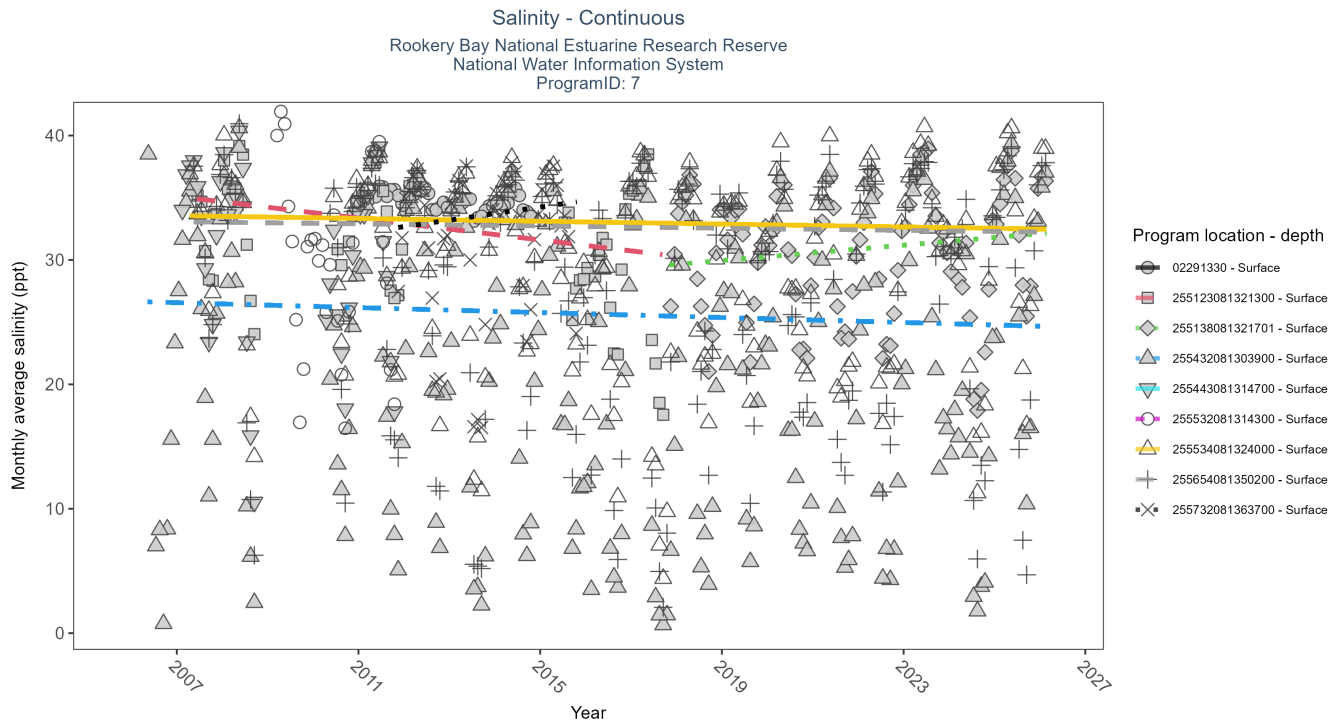


Figure 31: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 35: Seasonal Kendall-Tau Results for Salinity - Program 7

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
255123081321300	Significantly decreasing trend	1809	8	2007 - 2017	32	-0.23	35.11	-0.44	0.02
255138081321701	Significantly increasing trend	2986	10	2017 - 2026	32	0.17	29.33	0.31	0.03
255432081303900	No significant trend	6468	20	2006 - 2026	22	-0.08	26.67	-0.1	0.11
255443081314700	Insufficient data to calculate trend	1465	4	2007 - 2011	32	-	-	-	-
255532081314300	Insufficient data to calculate trend	902	3	2009 - 2011	31	-	-	-	-
255534081324000	No significant trend	6162	19	2007 - 2026	32	-0.04	33.56	-0.06	0.36
255654081350200	No significant trend	6183	19	2007 - 2026	32	-0.04	33.07	-0.04	0.36
255732081363700	No significant trend	1434	5	2011 - 2015	34	0.25	32.15	0.52	0.1
02291330	Insufficient data to calculate trend	1697	4	2011 - 2014	35	-	-	-	-

At two program locations, monthly average salinity increased by 0.31 ppt per year at one site and by 2.06 ppt per year at the other. At three program locations, monthly average salinity decreased between 0.06 and 0.44 ppt per year. No detectable change in monthly average salinity was observed at seven locations. There was insufficient data to fit a model for four locations.

Salinity - Continuous - Program 354

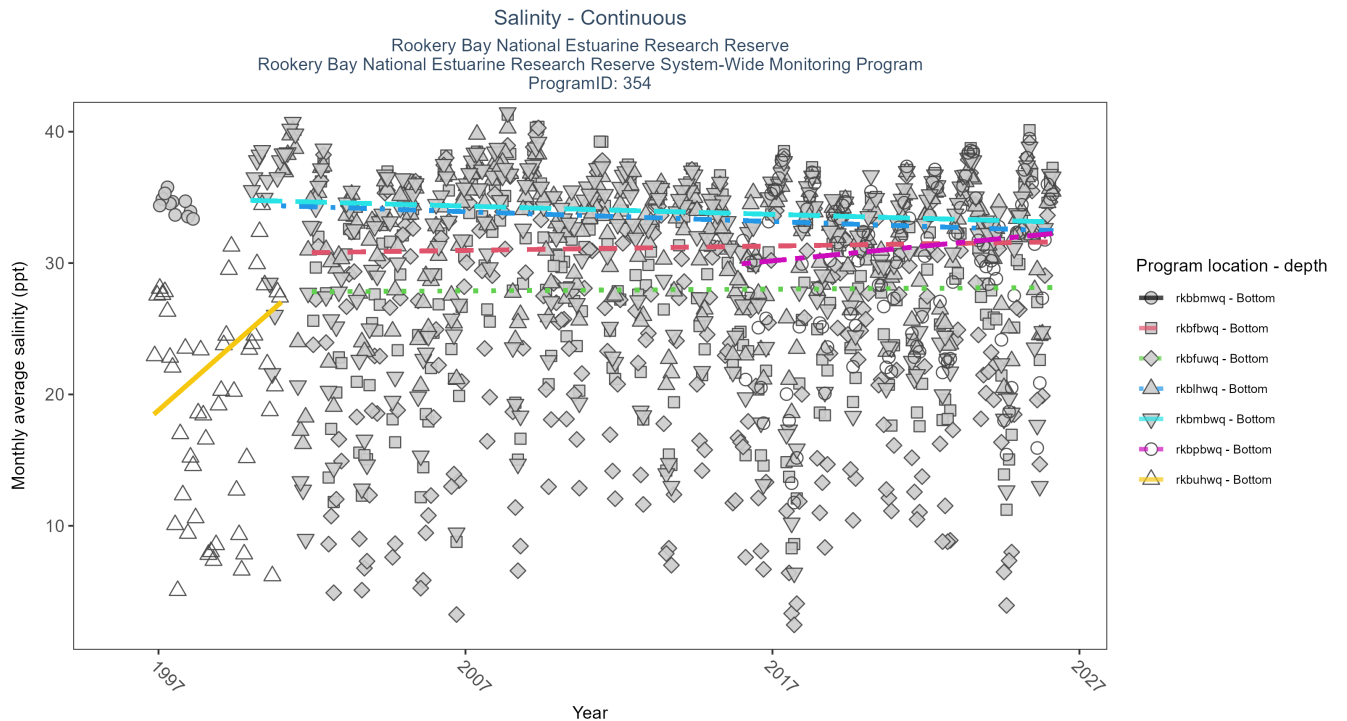


Figure 32: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 36: Seasonal Kendall-Tau Results for Salinity - Program 354

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfuwq	No significant trend	703182	25	2002 - 2026	29.9	0.04	30.8	0.03	0.34
rkbfuwq	No significant trend	717069	25	2002 - 2026	26.3	0.01	27.82	0.01	0.79
rkmbbwq	Significantly decreasing trend	736129	27	2000 - 2026	33.3	-0.1	34.78	-0.06	0.02
rkbpbwq	No significant trend	333127	11	2016 - 2026	30.5	0.14	29.94	0.23	0.06
rkbbmwq	Insufficient data to calculate trend	12256	2	1997 - 1998	34.5	-	-	-	-
rkblhwq	Significantly decreasing trend	700367	26	2001 - 2026	33.2	-0.13	34.36	-0.08	0
rkbuhwq	Significantly increasing trend	68446	5	1996 - 2000	21.1	0.31	16.74	2.06	0.04

At two program locations, monthly average salinity increased by 0.31 ppt per year at one site and by 2.06 ppt per year at the other. At three program locations, monthly average salinity decreased between 0.06 and 0.44 ppt per year. No detectable change in monthly average salinity was observed at seven locations. There was insufficient data to fit a model for four locations.

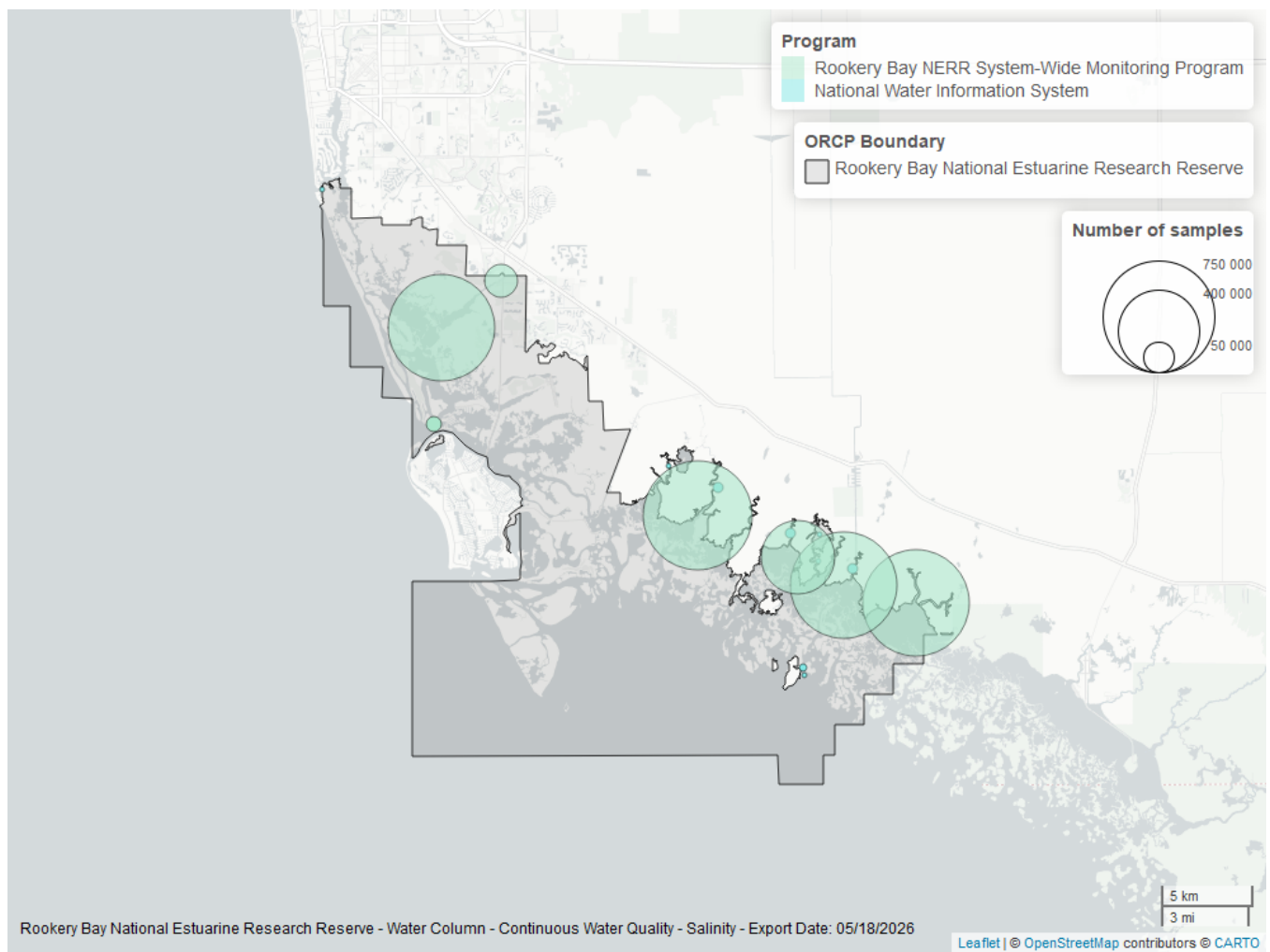


Figure 33: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous - Program 7

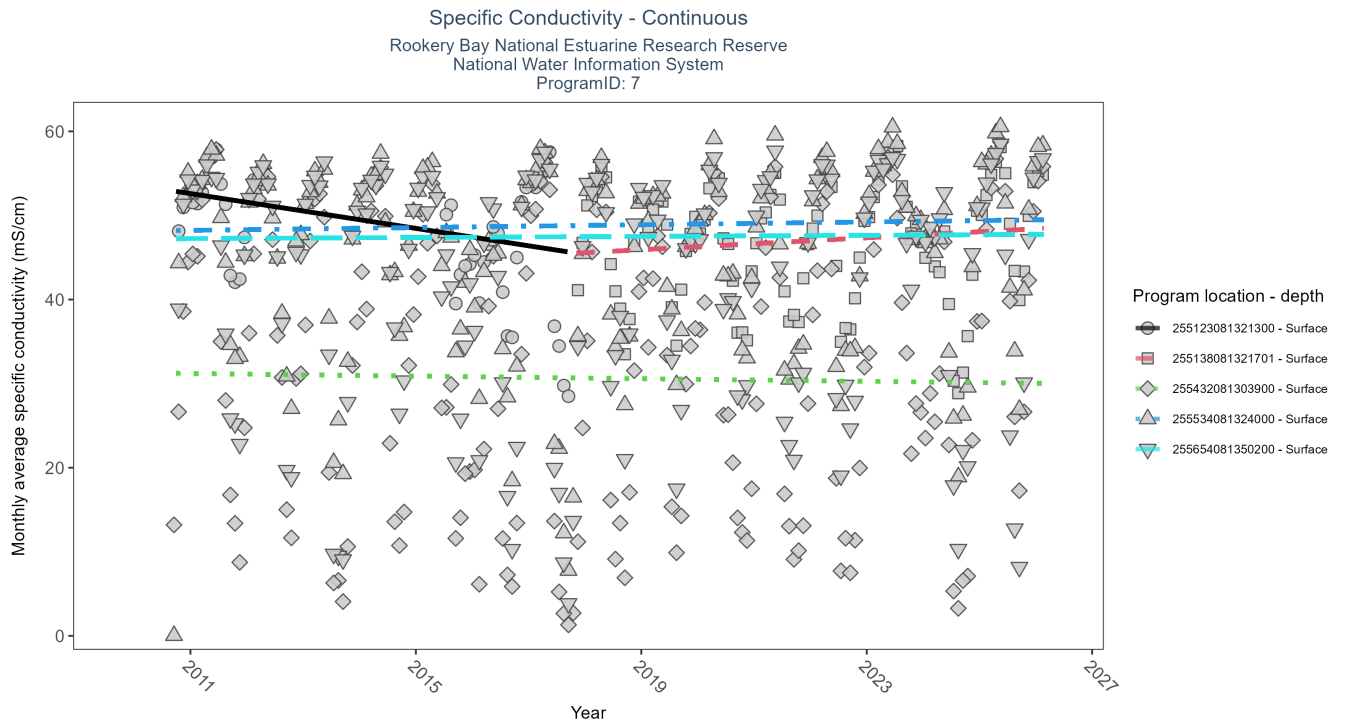


Figure 34: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 37: Seasonal Kendall-Tau Results for Specific Conductivity - Program 7

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
255123081321300	No significant trend	1295	6	2010 - 2017	49.1	-0.20	53.63	-1.03	0.20
255138081321701	Significantly increasing trend	2986	10	2017 - 2026	48.3	0.18	45.21	0.36	0.02
255432081303900	No significant trend	5512	17	2010 - 2026	34.4	-0.03	31.28	-0.08	0.61
255534081324000	No significant trend	5509	17	2010 - 2026	48.6	0.04	48.14	0.08	0.43
255654081350200	No significant trend	5547	17	2010 - 2026	48.6	0.03	47.21	0.03	0.56

At one program location, monthly average specific conductivity increased by 0.36 mS/cm per year. At two program locations, monthly average specific conductivity decreased by 0.08 mS/cm per year at one site and by 0.1 mS/cm per year at the other. No detectable change in monthly average specific conductivity was observed at eight locations. There was insufficient data to fit a model for one location.

Specific Conductivity - Continuous - Program 354

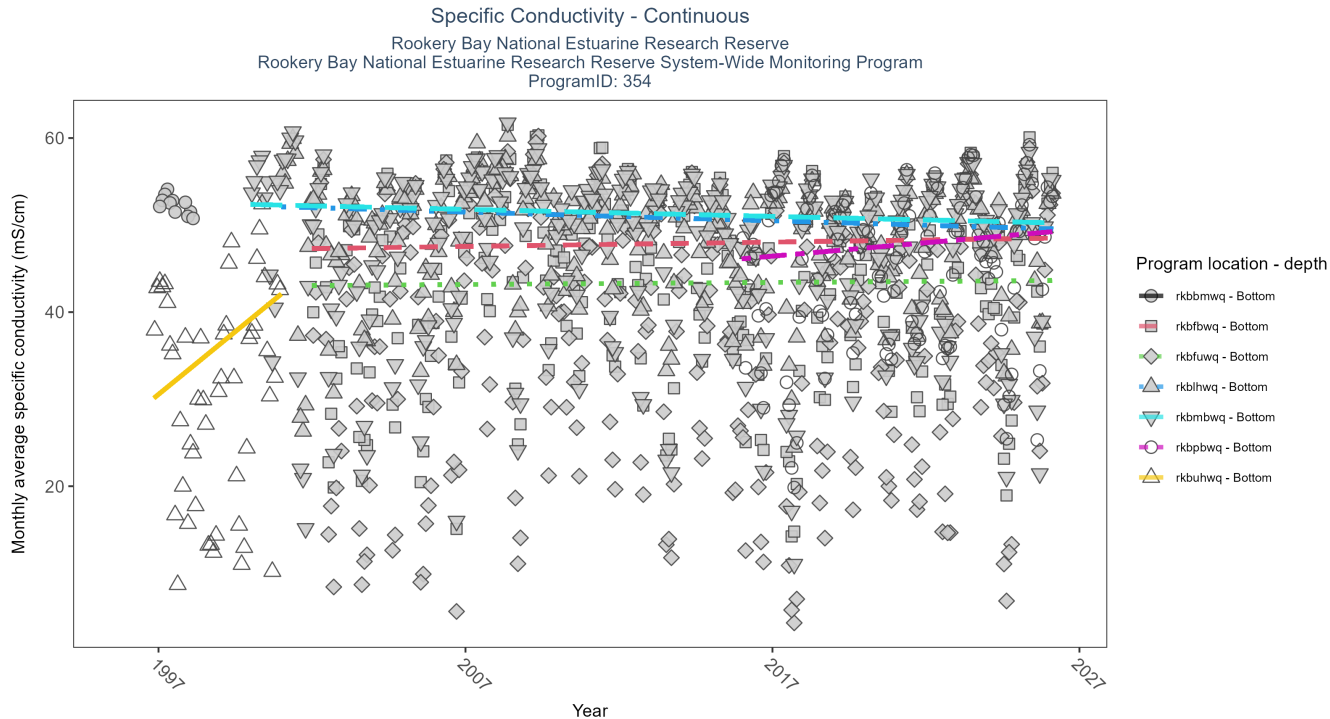


Figure 35: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 38: Seasonal Kendall-Tau Results for Specific Conductivity - Program 354

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfbwq	No significant trend	703174	25	2002 - 2026	46.07	0.04	47.3	0.05	0.33
rkbfuwq	No significant trend	717061	25	2002 - 2026	41.17	0.01	43.06	0.02	0.77
rkmbbwq	Significantly decreasing trend	736129	27	2000 - 2026	50.83	-0.09	52.37	-0.08	0.02
rkbpbwq	No significant trend	333114	11	2016 - 2026	47.03	0.13	46.12	0.31	0.07
rkbbmwq	Insufficient data to calculate trend	12256	2	1997 - 1998	52.41	-	-	-	-
rkblhwq	Significantly decreasing trend	700390	26	2001 - 2026	50.59	-0.12	52.11	-0.1	0
rkbuhwq	No significant trend	68971	5	1996 - 2000	33.92	0.29	27.67	2.88	0.06

At one program location, monthly average specific conductivity increased by 0.36 mS/cm per year. At two program locations, monthly average specific conductivity decreased by 0.08 mS/cm per year at one site and by 0.1 mS/cm per year at the other. No detectable change in monthly average specific conductivity was observed at eight locations. There was insufficient data to fit a model for one location.

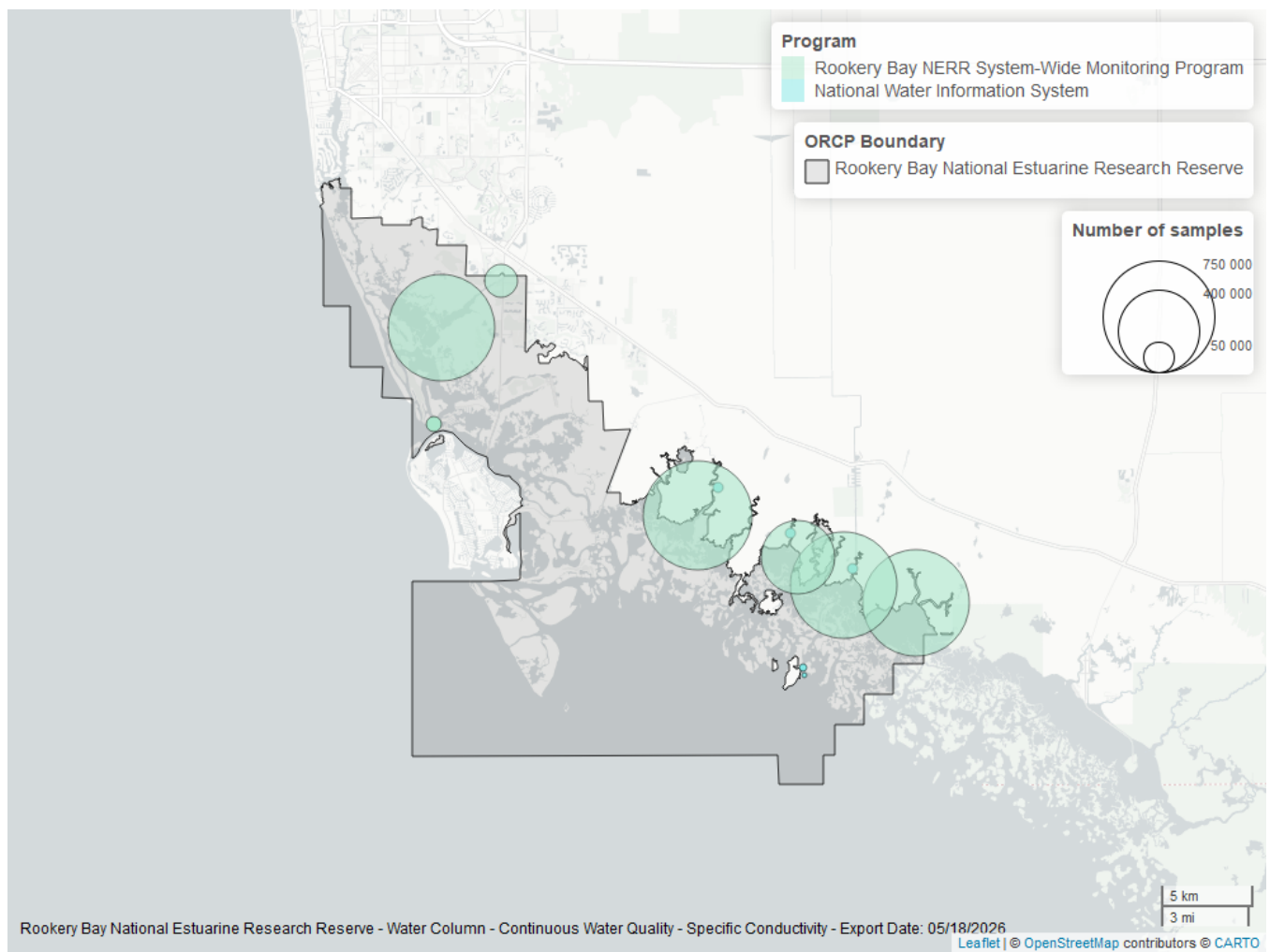


Figure 36: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Submerged Aquatic Vegetation

The data file used is: **All_SAV_Parameters-2026-Jun-04.txt**

Submerged aquatic vegetation (SAV) refers to plants and plant-like macroalgae species that live entirely underwater. The two primary categories of SAV inhabiting Florida estuaries are *benthic macroalgae* and *seagrasses*. They often grow together in dense beds or meadows that carpet the seafloor. *Macroalgae* include multicellular species of green, red and brown algae that often live attached to the substrate by a holdfast. They tend to grow quickly and can tolerate relatively high nutrient levels, making them a threat to seagrasses and other benthic habitats in areas with poor water quality. In contrast, *seagrasses* are grass-like, vascular, flowering plants that are attached to the seafloor by extensive root systems. *Seagrasses* occur throughout the coastal areas of Florida, including protected bays and lagoons as well as deeper offshore waters on the continental shelf. *Seagrasses* have taken advantage of the broad, shallow shelf and clear water to produce two of the most extensive seagrass beds anywhere in continental North America.

Parameters

Percent Cover measures the fraction of an area of seafloor that is covered by SAV, usually estimated by evaluating multiple small areas of seafloor. Percent cover is often estimated for total SAV, individual types of vegetation (seagrass, attached algae, drift algae) and individual species.

Frequency of Occurrence was calculated as the number of times a taxon was observed in a year divided by the number of sampling events, multiplied by 100. Analysis is conducted at the quadrat level and is inclusive of all quadrats (i.e., quadrats evaluated using Braun-Blanquet, modified Braun-Blanquet, and percent cover.”

Species

Turtle grass (*Thalassia testudinum*) is the largest of the Florida seagrasses, with longer, thicker blades and deeper root structures than any of the other seagrasses. It is considered a climax seagrass species.

Shoal grass (*Halodule wrightii*) is an early colonizer of vegetated areas and usually grows in water too shallow for other species except *widgeon grass*. It can often tolerate larger salinity ranges than other seagrass species. *Shoal grass* is characterized by thin, flat blades, that are narrower than *turtle grass* blades.

Manatee grass (*Syringodium filiforme*) is easily recognizable because its leaves are thin and cylindrical instead of the flat, ribbon-like form shared by many other seagrass species. The leaves can grow up to half a meter in length. *Manatee grass* is usually found in mixed seagrass beds or small, dense monospecific patches.

Widgeon grass (*Ruppia maritima*) grows in both fresh and salt water and is widely distributed throughout Florida's estuaries in less saline areas, particularly in inlets along the east coast. This species resembles *shoal grass* in certain environments but can be identified by the pointed tips of its leaves.

Three species of *Halophila* spp. are found in Florida - **Star grass** (*Halophila engelmannii*), **Paddle grass** (*Halophila decipiens*), and **Johnson's seagrass** (*Halophila johnsonii*). These are smaller, more fragile seagrasses than other Florida species and are considered ephemeral. They grow along a single long rhizome, with short blades. These species are not well-studied, although surveys are underway to define their ecological roles.

Notes

Star grass, *Paddle grass*, and *Johnson's seagrass* will be grouped together and listed as **Halophila spp.** in the following managed areas. This is because several surveys did not specify to the species level:

- Banana River Aquatic Preserve
- Indian River-Malabar to Vero Beach Aquatic Preserve
- Indian River-Vero Beach to Ft. Pierce Aquatic Preserve
- Jensen Beach to Jupiter Inlet Aquatic Preserve
- Loxahatchee River-Lake Worth Creek Aquatic Preserve
- Mosquito Lagoon Aquatic Preserve

- Biscayne Bay Aquatic Preserve
- Florida Keys National Marine Sanctuary

Rookery Bay National Estuarine Research Reserve
SAV Percent Cover - Sample Locations

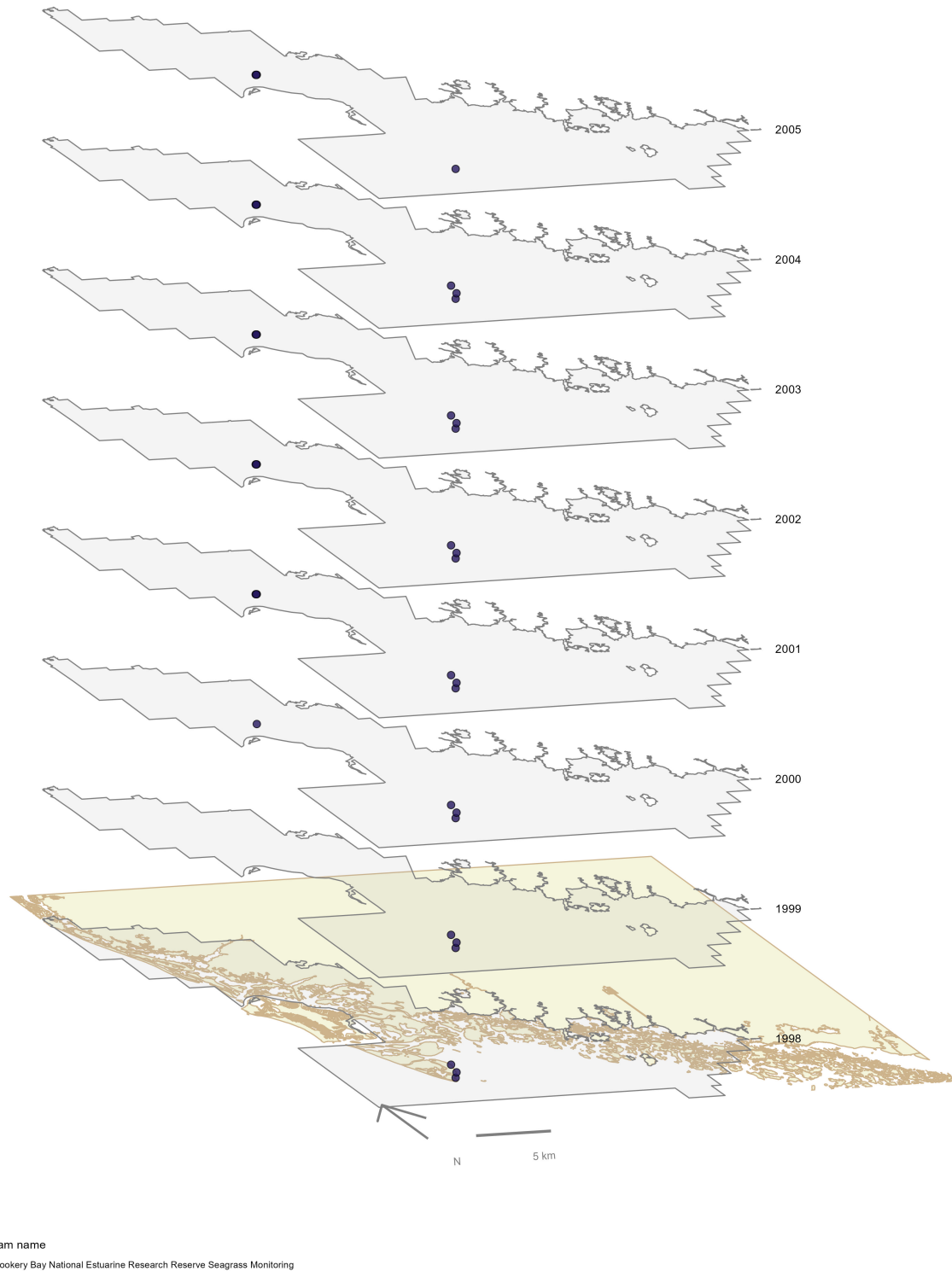


Figure 37: Maps showing the temporal scope of SAV sampling sites within the boundaries of *Rookery Bay National Estuarine Research Reserve* by program name.

Click [here](#) to view spatio-temporal plots on GitHub.

Sampling locations by Program:

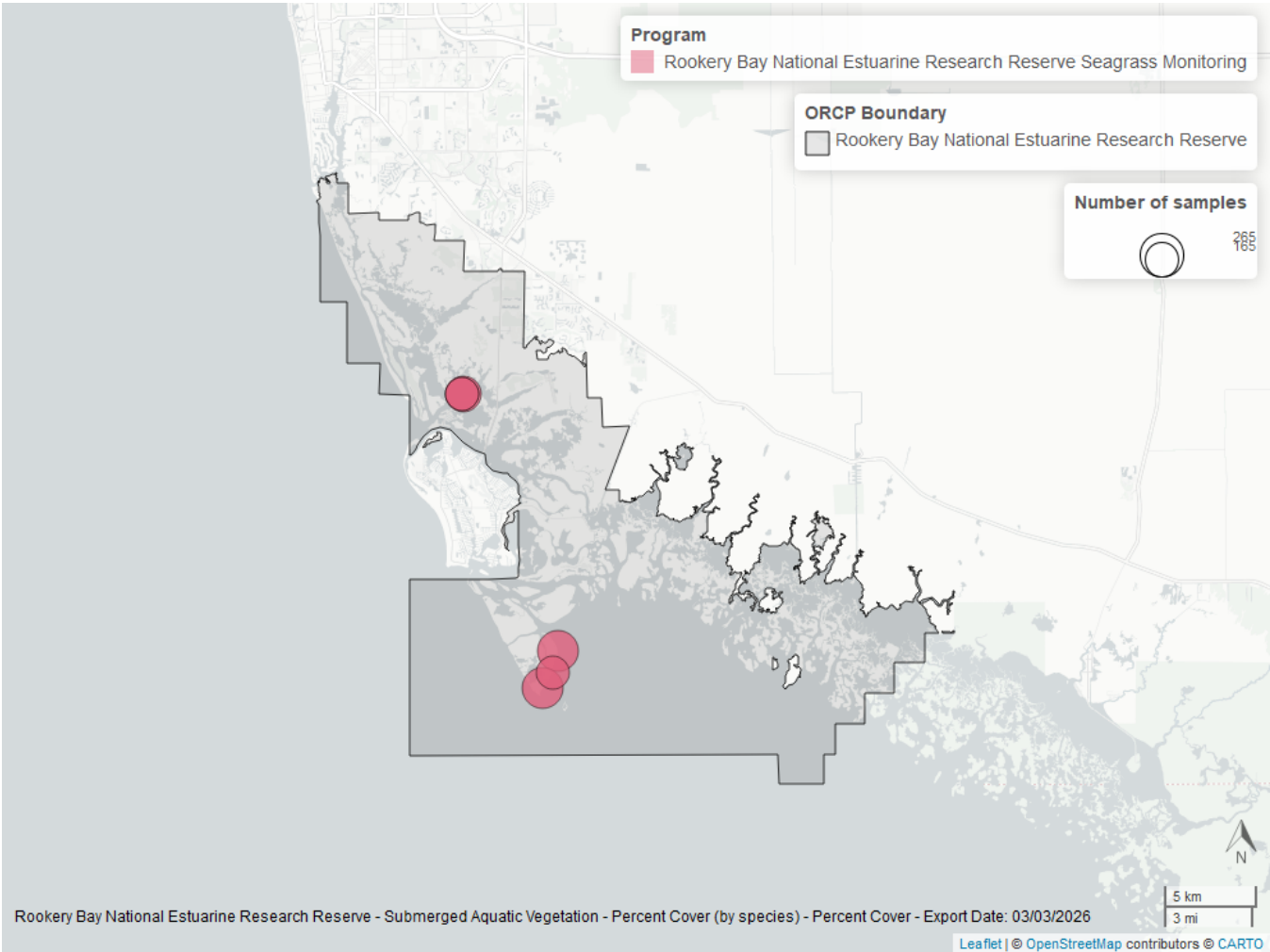


Figure 38: Map showing SAV sampling sites within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The point size reflects the number of samples at a given sampling site.

Table 39: Program Information for Submerged Aquatic Vegetation

<i>ProgramID</i>	<i>N-Data</i>	<i>YearMin</i>	<i>YearMax</i>	<i>method</i>	<i>Sample Locations</i>
572	1220	1998	2005	Percent Cover	6

Program names:

[572](#) - Rookery Bay National Estuarine Research Reserve Seagrass Monitoring¹⁰

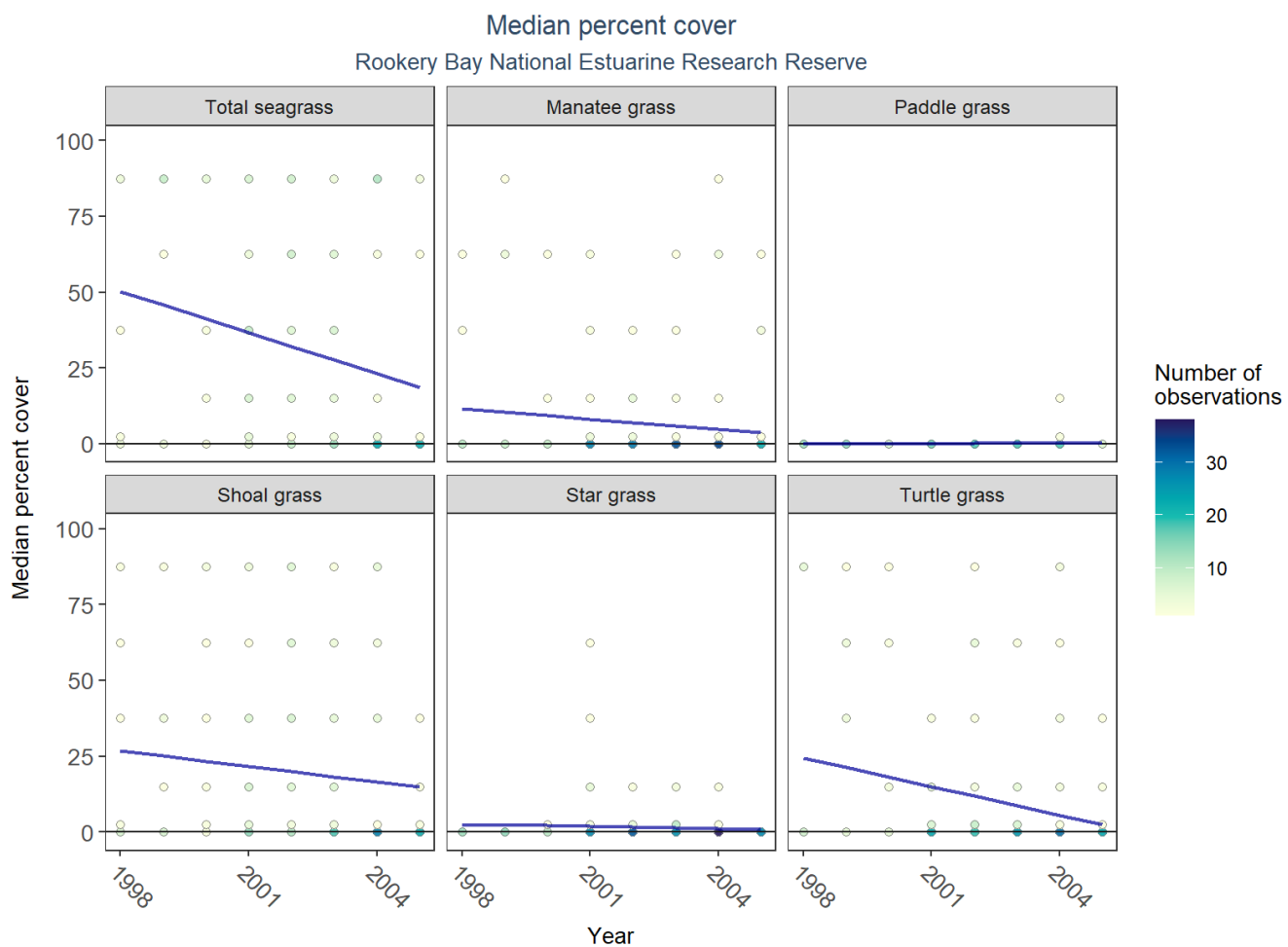


Figure 39: Scatter plots of median percent cover of submerged aquatic vegetation over time by group. Plots for time series that included five or more years of observations show the estimated trend as a blue line.

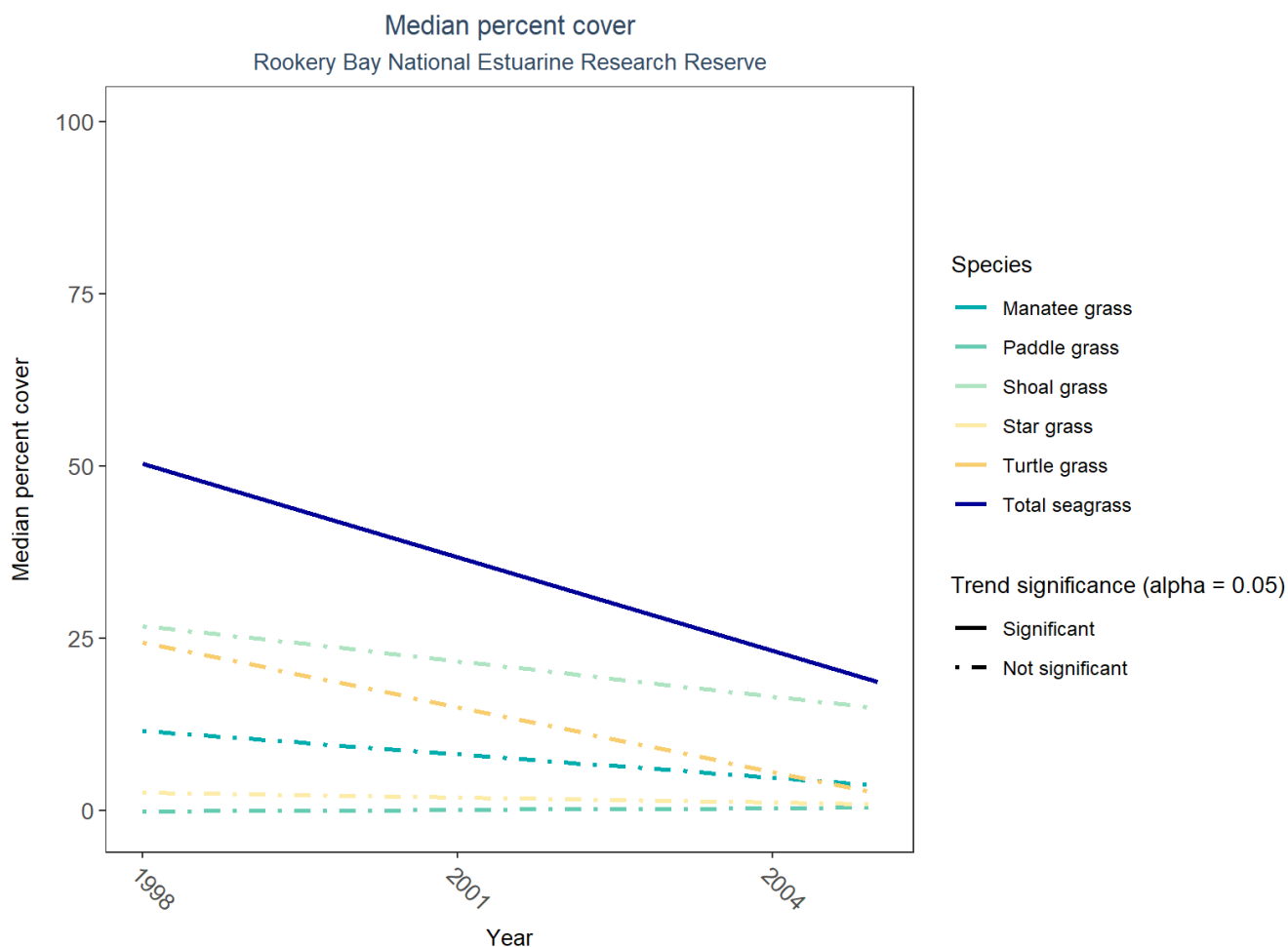


Figure 40: Trend lines of median percent cover of submerged aquatic vegetation over time for species that had five or more years of observations. Line type represents significance (solid) or non-significance (dashed) of the trends.

Table 40: Percent Cover Trend Analysis for Rookery Bay National Estuarine Research Reserve

<i>Species</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>LME Intercept</i>	<i>LME Slope</i>	<i>P</i>
Shoal grass	No detectable trend	1998 - 2005	33.6286656	-1.7090982	0.2546051
Paddle grass	No detectable trend	1998 - 2005	-0.4445673	0.0784252	0.3562621
Star grass	No detectable trend	1998 - 2005	3.5304851	-0.2323267	0.4437117
Manatee grass	No detectable trend	1998 - 2005	16.1047805	-1.1343837	0.4000856
Turtle grass	No detectable trend	1998 - 2005	36.9900160	-3.1381459	0.0666593
Total seagrass	Decreasing trend	1998 - 2005	68.3692745	-4.5209435	0.0166191

An annual decrease in percent cover was observed for total seagrass (-4.52%). Manatee grass, paddle grass, shoal grass, star grass, and turtle grass showed no detectable change in percent cover.

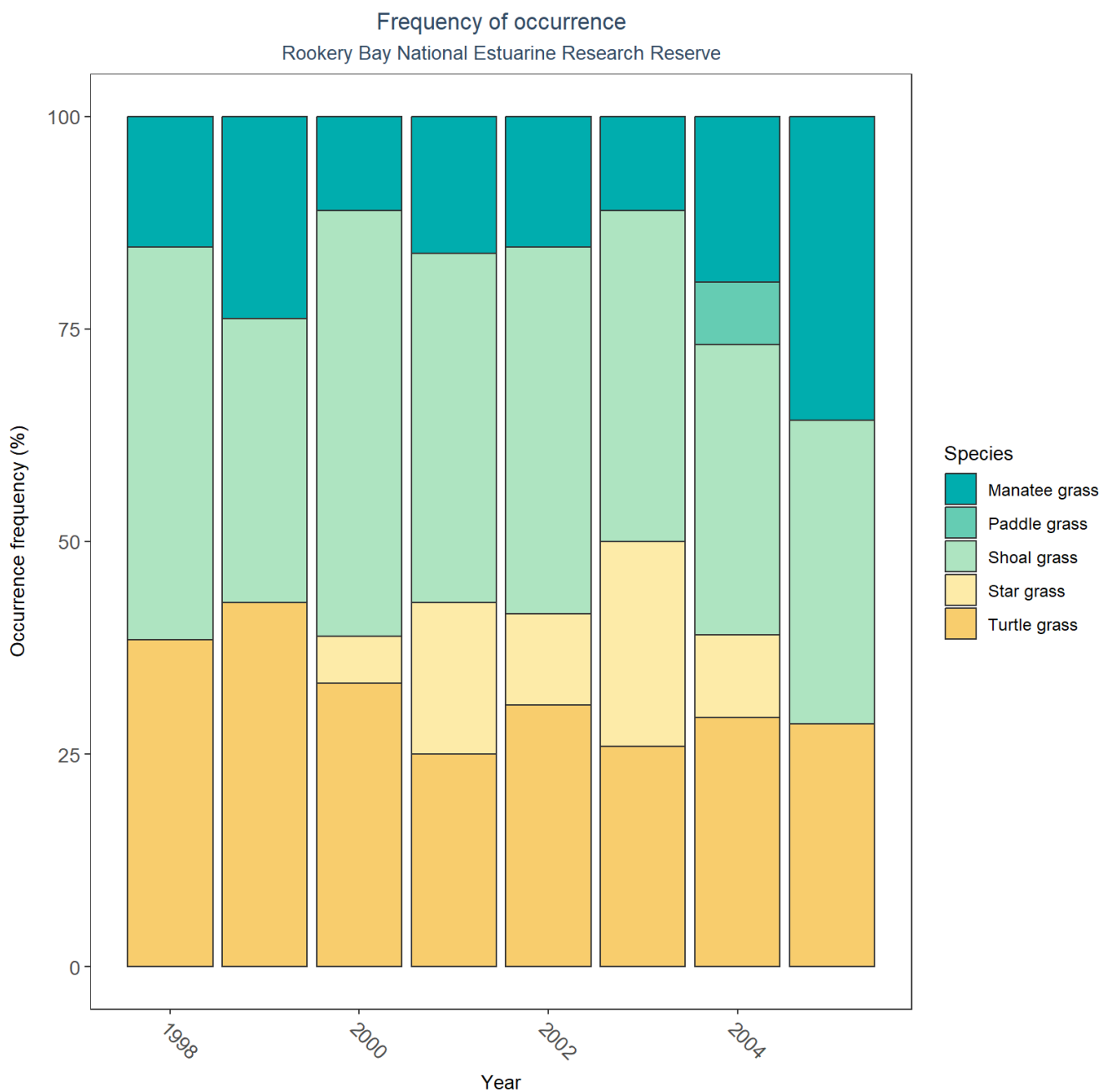


Figure 41: Bar plot of submerged aquatic vegetation species occurrence frequency over time by group.

SAV Water Column Analysis

The following parameters are available for Rookery Bay National Estuarine Research Reserve within the SAV_WC_Report:

- Colored Dissolved Organic Matter
- Chlorophyll a
- Dissolved Oxygen
- Dissolved Oxygen Saturation
- pH

- Salinity
- Secchi Depth
- Water Temperature
- Total Nitrogen
- Total Suspended Solids
- Turbidity

Access the reports here: [DRAFT_SAV_WC_Report.pdf](#)

Nekton

The data file used is: **All_NEKTON_Parameters-2026-Jun-04.txt**

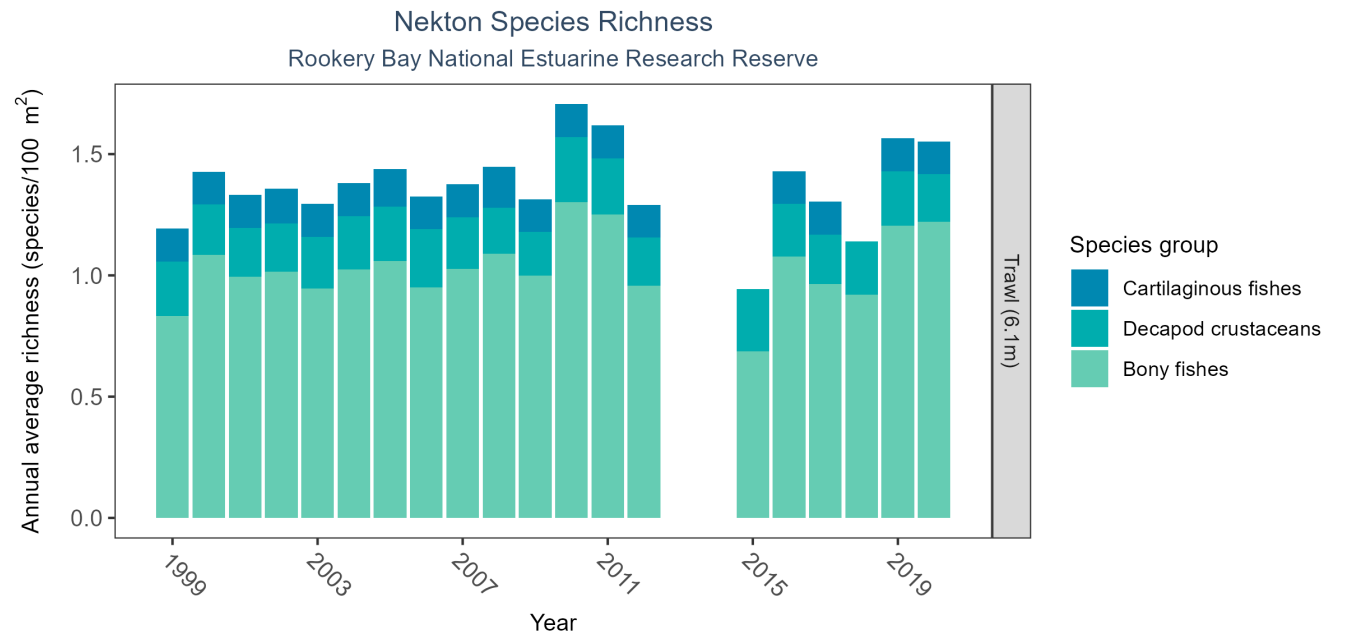


Figure 42: Bar graph(s) of annual average nekton richness over time for species groups occurring in at least 1% of samples. The bar colors represent species groups including bony fishes, cartilaginous fishes, decapod crustaceans (e.g., shrimps, crabs, and lobsters), and cephalopods (e.g., squid). Gear types and sizes are indicated in the panel label.

Table 41: Nekton Species Richness

<i>Gear Type</i>	<i>No. of Samples</i>	<i>No. of Years with Data</i>	<i>Period of Record</i>	<i>Median No. of Taxa</i>	<i>Mean No. of Taxa</i>
Trawl (6.1 m)	6071	20	1999 - 2020	0.27	0.63

The median annual number of taxa was 0.27 based on 6,071 observations collected by 6.1-meter trawl between 1999 and 2020.

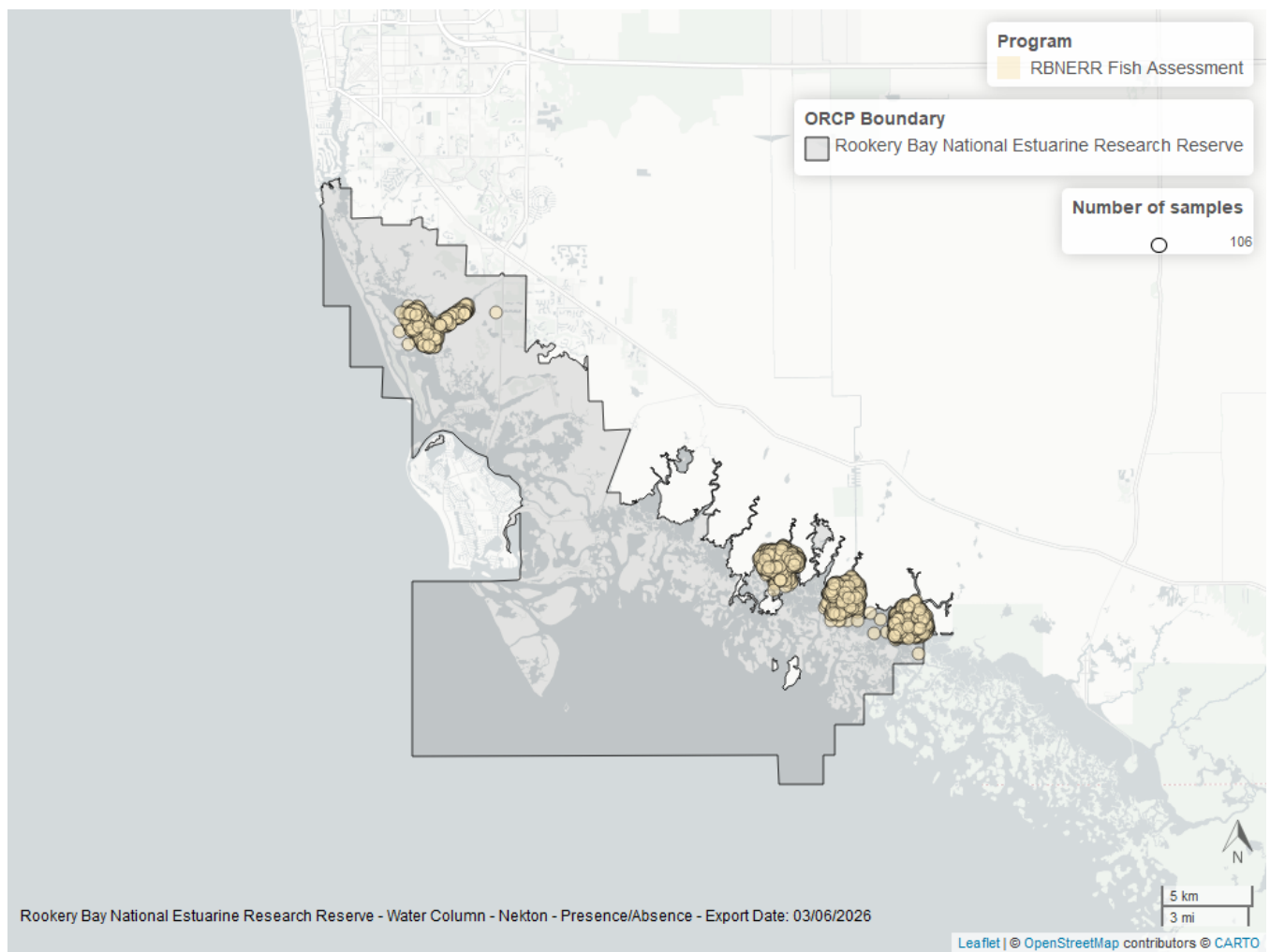


Figure 43: Map showing location of nekton sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Species list

Acanthostracion quadricornis ²	Eugerres plumieri ²	Mugil spp. ²
Achirus lineatus ²	Floridichthys carpio ²	Mycteroperca microlepis ²
Albula vulpes ²	Ginglymostoma cirratum ²	Myrophis punctatus ²
Aluterus schoepfii ²	Gobiesox strumosus ²	Nicholsina usta ²
Anarchopterus criniger ²	Gobionellus oceanicus ²	Ogcocephalus cubifrons ²
Anchoa spp. ²	Gobiosoma bosc ²	Ogilbia cayorum ²
Ancylopsetta quadrocellata ²	Gobiosoma robustum ²	Oligoplites saurus ²
Archosargus probatocephalus ²	Gunterichthys longipenis ²	Opisthonema oglinum ²
Archosargus rhomboidalis ²	Gymnura micrura ²	Opsanus beta ²
Ariopsis felis ²	Halodule wrightii ¹	Orthopristis chrysoptera ²
Bagre marinus ²	Halophila decipiens ¹	Paralichthys albigutta ²
Bairdiella chrysoura ²	Halophila engelmannii ¹	Penaeus spp. ²
Brevoortia smithi ²	Harengula jaguana ²	Pogonias cromis ²
Brevoortia spp. ²	Hippocampus erectus ²	Prionotus scitulus ²
Calamus arctifrons ²	Hippocampus zosterae ²	Prionotus tribulus ²
Callinectes sapidus ²	Hoplosternum littorale ²	Rachycentron canadum ²
Caranx spp. ²	Hypanus americanus ²	Rhinoptera bonasus ²
Caulerpa spp. ¹	Hypanus sabinus ²	Sciaenops ocellatus ²
Centropomus undecimalis ²	Hyporhamphus unifasciatus ²	Scorpaena brasiliensis ²
Chaetodipterus faber ²	Hypostomus plecostomus ²	Selene vomer ²
Chasmodes saburrae ²	Hypsoblennius hentz ²	Serraniculus pumilio ²
Chilomycterus schoepfii ²	Lagodon rhomboides ²	Sphoeroides nephelus ²
Chloroscombrus chrysurus ²	Leiostomus xanthurus ²	Sphoeroides spengleri ²
Cichlidae spp. ²	Lepomis spp. ²	Sphyraena guachancho ²
Citharichthys macrops ²	Leptocephalus larvae ²	Stephanolepis hispidus ²
Citharichthys spilopterus ²	Lophogobius cyprinoides ²	Symphurus plagiusa ²
Clupeidae ²	Lucania parva ²	Syngnathus louisianae ²
Ctenogobius boleosoma ²	Lutjanus analis ²	Syngnathus scovelli ²
Ctenogobius smaragdus ²	Lutjanus griseus ²	Synodus foetens ²
Cynoscion arenarius ²	Lutjanus synagris ²	Syringodium filiforme ¹
Cynoscion nebulosus ²	Menippe mercenaria ²	Thalassia testudinum ¹
Diplectrum formosum ²	Menticirrhus spp. ²	Total seagrass ¹
Echeneis neucratoides ²	Microgobius gulosus ²	Trachinotus falcatus ²
Elops saurus ²	Microgobius thalassinus ²	Trinectes maculatus ²
Epinephelus itajara ²	Micropterus salmoides ²	Urophycis floridana ²
Etropus crossotus ²	Monacanthus ciliatus ²	Acanthostracion quadricornis ²
Eucinostomus spp. ²	Mugil cephalus ²	Achirus lineatus ²

1 - Submerged Aquatic Vegetation, 2 - Nekton

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